

Effect of exogenous market sentiment indicators in stock price direction prediction

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ARTICLE INFO

Keywords:

XGB Classifier
Technical Indicators
Best Feature Subset Selection
Exogenous Market Indicators
VIX
Market Volatility Index
Fear-Greed Index
Put-Call Ratio
Collective Behaviour

ABSTRACT

As the most popular methodology in stock price prediction, technical analysis accounts for 66 % of stock price prediction studies for 11 years' period (2007 ~ 2018), meanwhile fundamental analysis and combined analysis using technical and fundamental analysis are just 23 % and 11 % of studies, respectively. Technical stock price analysis generally expands the feature space by generating price-derived technical indicators from daily stock prices. However, using price-derived technical indicators is criticized for its complete subjectivity and self-fulfilling prophecy. As a remedial action for price-derived technical indicators, this study includes three popular exogenous market sentiment indicators (EMIs): market volatility index (VIX), fear-greed index (FGX), and put-call ratio (PCR), and investigates the importance and effect of the selected EMIs in stock price direction prediction. To capture the maximal effect of the EMIs in stock price direction prediction, the best feature subset (BFS) selection method is adopted in company with the extreme gradient boosting (XGB) algorithm, which significantly improves the prediction accuracy and interpretability with fewer input features. The proposed BFS with XGB method adequately reflects the collective behaviour of the EMIs with price-derived technical indicators of time series stock price data. Additionally, combined with a piecewise time segmentation technique, the piecewise best feature subset (PBFS) selection method shows that the three EMIs: VIX, FGX, and PCR make a considerable effect in prediction accuracy and interpretation of stock price direction prediction.

1. Introduction

Recently public sentiment analysis has attracted growing interest in the stock market prediction study. The public sentiment analysis gathers public opinions, emotions, and feelings about the stock market, economy, or companies from such social media as X (formerly known as Twitter) and Meta (formerly known as Facebook) or various finance news websites. Once web-scraped public opinions are collected, the gathered data are transformed into quantified sentiment scores by using Natural Language Processing (NLP) tools such as Natural Language Tool Kit (NLTK), Valence Aware Dictionary for Sentiment Reasoning (VADER), SentiWordNet, or Bidirectional Encoder Representations from Transformers (BERT) in terms of positive, negative, or neutral sentiments (Das et al., 2022). Then, ML or DL algorithms process the sentiment data for the prediction of future stock prices. However, the predictive power of NLP-based sentiment analysis varies over input datasets, input features, and prediction algorithms. Ranco et al. (2015) showed that there is a weak or not strong Pearson correlation and Granger causality between Twitter sentiment and stock prices, and the

dependence between them is useful only when data are properly selected, or different sources of data are analyzed together. Deveikyte et al. (2022) also showed that many sentiments score-based stock market prediction studies do not achieve satisfactory prediction accuracy. Still, there is no clear evidence for a significant cause-and-effect relationship between sentiment scores of public opinions and stock market price movements regardless of market conditions (Smith & O'Hare, 2022). Accordingly, the prediction accuracy of NLP-based public sentiment stock price analysis crucially depends on the quality, scope, and relationship of input features, although some correlations between stock prices and sentiment scores have been disclosed.

Another method of stock price prediction study is fundamental stock price analysis. In fundamental analysis, stock market price is considered to be subject to its intrinsic value or equilibrium price. Stock's intrinsic value and the earning potential of the stock is determined by periodic financial statements such as balance sheets, income statements, and cash flow statements, as well as managerial status, industry outlook, and economic situation and prospect (Murphy, 1999). However, measuring the fair market value of a stock based on such external sources as

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<https://doi.org/10.1016/j.eswa.2025.127696>

Received 28 June 2024; Received in revised form 17 March 2025; Accepted 10 April 2025

Available online 15 April 2025

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financial statements, company performance, and economic environment is not only difficult, but also vulnerable to intentional or unintentional human errors. In addition, fundamental analysis does not capture the timely sensitive stock price changes. A company's monthly, quarterly, or annual financial statements and reports do not properly reflect momentarily changing stock prices (Yun et al., 2021).

Meanwhile, technical stock price analysis uses historical stock prices and trading volume to create price-derived technical indicators and identifies recurring patterns, trends, and strengths within the set of historical prices and technical indicators to predict price changes (Murphy, 1999). Technical analysis not only can reflect constantly and unexpectedly changing stock prices, but also can avoid such inevitable drawbacks of fundamental analysis as human interventions, errors, or fraud. For this reason, numerous previous studies have shown the popularity and effectiveness of technical analysis. According to previous studies, 87 % of 692 surveyed fund managers think technical analysis important for their investment decisions (Menkhoff, 2010; Fang et al., 2014), and 56 out of 95 technical analysis produces supportive profitability (Park & Irwin, 2007; Fang et al., 2014).

Technical analysis generally expands the feature space by generating price-derived technical indicators from the daily stock prices. However, using price-derived technical indicators is criticized for 'complete subjectivity' and 'self-fulfilling prophecy' (Murphy, 1999) even with advantages of using technical indicators: effectiveness, simplicity and time-sensitivity for price prediction (Suzuki & Ohkura, 2016). Hence, using price-derived technical indicators is sometimes downgraded as pseudoscience, and is considered to cause data snooping and sample bias problems (Paulos, 2007; Fang et al., 2014).

As a solution for the problems of internal price-derived technical indicators, adopting exogenous market indicators (EMIs) such as advance and decline lines, the Arms index, and volatility index (VIX) expands the information beyond limited price and volume data to more persuasive financial information (Fang et al., 2014). There are two types of technical market indicators: market sentiment indicators and market strength indicators. The former predicts price movements based on the bullish or bearish psychology of the market. When bullish sentiment dominates the market, prices rise, otherwise, prices decline. Meanwhile, market strength indicators use the strength or breadth of market movements. The stronger the market moves, the longer the higher or lower price continues (Fang et al., 2014). There have been many studies that investigated the relationship between EMIs and stock price changes. Among the technical market indicators, the market volatility index (VIX) is one of the most popular indicators. VIX is generally taken as the barometer of the whole market, let alone the stock market. Accordingly, numerous studies use VIX to predict market volatility and achieve good prediction results (Chen et al., 2021). Along with VIX, fear-greed index (FGX) and put-call ratio (PCR) are also popular EMIs among investors and fund managers in the stock price prediction. Exogenous market indicators are the same time series data as historical stock price values, and they can add 'significant depth to technical analysis, because they contain much more information than price and volume' (Achelis, 2001; Fang et al., 2014).

However, to the best of our knowledge, no previous research investigated the importance and effect of EMIs together with price-derived technical indicators in stock price prediction. The adoption of EMIs can relieve the drawbacks of using price-derived technical indicators alone in stock price prediction. Additionally, the combination of input feature set of EMIs and price-derived technical indicators can make better interpretability and predictive power than NLP-based sentiment stock price analysis which is still questionable regarding its effectiveness.

To investigate the importance and effect of EMIs along with price-derived technical indicators in stock price movement direction prediction, the BFS selection method is used with the extreme gradient boosting (XGB) classification algorithm. The BFS selection method was proposed in the previous study (Yun et al., 2023), and it provided a good

prediction performance and interpretability by capturing collective behavior of input features in time series stock price data. Collective behavior or collectivity of input features is a property by which learning performance and prediction accuracy are changed depending on the set of input features. Furthermore, the BFS also effectively delivers the temporality of time series stock price data when it is used with piecewise time segmentation and smoothing of time series stock price data. Temporality is a property by which learning performance and prediction accuracy of stock price data depend on timeline as the functional value of time. Consequently, the piecewise best feature subset (PBFS) method, which is the BFS combined with a piecewise time segmentation and smoothing of time series data, can produce preferable local interpretability and predictability. (Yun et al., 2023).

The contribution of this study in stock market prediction study is summarized as follows. This study successfully captures the importance and effect of the selected EMIs in stock price direction prediction. Sentiment analysis of stock price prediction receives growing attention recently, but the consensus about the credibility and effectiveness of public sentiment scores in stock price prediction is not yet established. In lieu of Natural Language Processing based sentiment stock price prediction, adopting important EMIs and utilizing them as input features makes simpler, flexible, and more reliable results in stock price prediction studies. Secondly, this study provides an empirical application of the BFS selection method to EMIs in the stock price prediction studies. To capture the maximal effect of the EMIs in stock price direction prediction, the BFS selection method is adopted in company with the XGB algorithm, which significantly improves the prediction accuracy and interpretability with fewer input features. The previous study (Yun et al., 2023) introduced the BFS method and experimented on the sole set of price-derived technical indicators for the validity of the methodology but did not show the sufficient applicability of the proposed methodology. This study extends the practicability of the best feature subset method to any external features including market indicators. Consequently, timely flexible local and global interpretability with better prediction accuracy to handle constantly changing stock prices are achievable in stock price prediction. Third, this study provides a frame to relieve the criticism about price-derived technical indicators: 'complete subjectivity' and 'self-fulfilling prophecy'. To this end, datasets are constructed by combining price-derived technical indicators and exogenous market indicators together, and the performance of exogenous market indicators are measured and compared to important price-derived indicators. Moreover, the inclusion of EMIs expands the interpretable power beyond historical stock price and volume data to a variety of financial and economic information.

The rest of this paper is organized as follows: In Section 2, related literature on market price prediction studies is reviewed. Section 3 describes the proposed stock price movement direction prediction method in detail. Section 4 discusses experimental results, Section 5 and Section 6 provides the comparison of the proposed method to previous stock price prediction studies and NLP-based sentiment analysis, respectively. Section 7 concludes the research.

2. Literature review

The share of technical analysis in the recent stock market price prediction studies becomes overwhelming. According to a survey about the stock market prediction studies published in academic journals over 11 years period (2007 ~ 2018), 66 % of research used technical analysis and only 23 % and 11 % were based on fundamental analysis and combined analysis, respectively (Nti et al., 2020). The popularity of technical analysis in stock price prediction study is substantially attributed to the rapid development and remarkable successes of Machine Learning (ML) and Deep Learning (DL) algorithms. ML and DL prediction models not only achieve a great degree of accuracy, but also handle datasets without checking such statistical data assumptions as linearity, independence, homoscedasticity, stationarity, and normality.

Various ML models such as support vector machine (SVM), Naïve Bayes learning, k-nearest neighbor, and tree-structured learning algorithms: decision trees, extra trees, random forest, and XGB along with DL models: Convolutional neural network (CNN), recurrent neural network (RNN), and long short term memory networks (LSTM) are adopted for stock market prediction studies over the past decade (Yun et al., 2021; Lin & Marques, 2024).

There are several ways to improve model's prediction performance in stock price prediction studies. Some ML and DL stock price prediction studies elaborate on feature engineering of technical indicators. In particular, technical analysis generally expands the feature space by generating various price-derived technical indicators as statistical functions of historical prices or trading volume. However, using price-derived technical indicators is criticized because they cause such problems as 'complete subjectivity' and 'self-fulfilling prophecy' (Murphy, 1999; Yun et al., 2023). That is, price-derived technical indicators do not provide any essential causation about future stock prices because they are just byproducts of past prices and volume.

Another type of technical indicator, exogenous market indicator, represents various financial market information such as investor sentiment, implied market volatility, and put option and call option relationship. Exogenous market indicators (EMIs) are very important to stock market practitioners but did not draw as much attention as price-derived technical indicators (Fang et al., 2014). Among them, the Implied Volatility (VIX) is one of the most frequently used exogenous market indicators. The VIX was developed in 1993 to measure the volatility of the S&P 500 index by the Chicago Board Options Exchange (CBOE). Since the creation of the VIX as a market volatility indicator, it has become popular because it is derived from the S&P 500 index options and is a forward-looking index of the expected return volatility of the S&P 500 index over the next 30 days (Whaley, 2009). It implies investors' expectations for the relative strength of price changes within a short period of the near future. Hence, as a well-known fear index for asset markets, it reflects both stock market uncertainty and a variance risk premium from selling stock market variance in a swap contract (Whaley, 2009; Bekaert & Hoerova, 2014). There have been many studies to analyze financial markets using VIX. Ruan (2018) explored the influence of VIX on the several counties' stock markets and found that VIX has a significant impact on the linkage between stock markets. Bardgett et al. (2019) analyzed the contribution of VIX to the option pricing model. They found that S&P 500 and VIX index returns, and option prices contain valuable information on the risk-neutral conditional distributions of volatility at different time horizons, which is not spanned by the S&P 500 market. Choi & Hong (2020) examined the causal relationships among the crude oil volatility index (OVX), VIX, and KOSPI 200 index volatilities (VKOSPI). In their study, OVX and VIX show bi-directional causality, and also OVX and VIX show causality to the VKOSPI. Chen et al. (2021) investigated whether investor sentiment has stronger predictive power than VIX and uncertainty indices in

predicting energy volatility. Consequently, either VIX or investor sentiment score does not have an absolutely stronger predictive power than the other. The appropriate balance and harmony of using both n features are quite valuable when predicting the volatility of crude oil-related assets.

However, the most recent financial market studies did not use VIX in stock price prediction. Table 1 shows recent financial market analysis studies employing EMIs. Only two studies out of nine studies in Table 1 have recently used EMIs including VIX in their stock price predictions. Qadan et al. (2024) examined how the shape of the VIX's term structure affects the evolution of the stocks' prices. Bouri et al. (2024) investigated the relationship between VIX and SKEW indices and the returns of the S&P 500 market indices. Other studies used EMIs for cryptocurrency market analysis or option pricing.

Similar to popular VIX, other exogenous market indicators used in this study: fear greed index (FGX) and put call ratio (PCR) are also not frequently adopted in stock price prediction studies. (Blau and Brough, 2015) examined the relationship between put-call ratios, short-selling activity, and constraints to short selling and their results showed that put-call ratios are inversely relation to proxies for short-sale constraints and negative relation between put-call ratios and future returns is orthogonal to proxies for short-sale constraints. Jena et al. (2019) examined the efficacy of the Put Call ratio in predicting market return at different time scales. Their findings show that appropriately using PCR makes an efficient predictor of the market return. Gang et al. (2020) investigated the influence of the put call ratio implied by the Shanghai Stock Exchange ETF option on the price discovery process of the SSE50 index. They suggested that the PCRs implemented in many trading practices may be misused, because there is no evidence that the PCRs and index returns are correlated. Meanwhile, this study adopts three EMIs: VIX, FGX, and PCR and investigates the importance and effect of the selected EMIs in stock price direction prediction.

Stock price and technical indicators are time series data in which input features make intricate interactions with each other along the timeline. To reflect these collective behaviors and effect change of the input features over time, the BFS selection method is adopted in company with the XGB algorithm. The BFS method was introduced in a previous study to capture the collectivity and temporality of stock price behavior (Yun et al., 2023).

The BFS selects the best performing subset of input features that can provide the best prediction accuracy and important feature based interpretation of the prediction. Furthermore, combined with piecewise segmentation of timeline, it produces more accurate local interpretability of fast changing stock price in a timely manner. To investigate the effect of exogenous market indicators among price-derived technical indicators, this study adopts the BFS method with three exogenous market indicators: VIX, FGX, and PCR. To predict the stock price movement direction, the XGB classifier is adopted as the main predictor in this study. A detailed description of the proposed methodology is

Table 1
Recent financial market analysis studies employing EMIs.

Authors	Year	Dataset	EMIs	Methods	Outcome
(Barua and Sharma, 2023)	2023	Global Market ETFs	FGX	CEEMDAN-GRU DL, XGB, Black-Litterman approach	Optimum stock portfolio
(Gaies et al., 2023)	2023	Bitcoin prices	CFGI	Granger causality test	Bitcoin price
(Wang et al., 2024)	2024	Cryptocurrencies	CFGI	Descriptive statistics	Cryptocurrencies price synchronicity
(Fang et al., 2024)	2024	SSE50, CSI300, CSI 500 index futures	VIX	Lasso, Gated Recurrent Unit	Prediction efficiency
(Escobar-Anel et al., 2024)	2024	Option prices	VIX	affine GARCH	Option pricing
(Qadan et al., 2024)	2024	5 US stocks	VIX	VIX's term structure	Stock price
(Bouri et al., 2024)	2024	S&P500, 11 sector idices	VIX, SKEW	Wavelet Localized Multiple Correlation	Market index returns
(Jacobs and Mai, 2024)	2024	Option prices	VIX, SPX options	autoregressive vector autoregressive (VAR) framework	Option pricing
(Felföldi-Szűcs and Králik, 2024)	2024	Cryptocurrencies	Put-Call Parity	zero arbitrage profit relationship	Cryptocurrency option pricing

provided in the next section.

3. Proposed methodology

3.1. Dynamics of stock price movement

Stock price is constantly changed by various internal and external driving forces. The sign of driving force of price change is either positive that it leads to price increase, or negative that it leads to price decrease. Specifically, P_{t+1} , stock price of day $t + 1$, is changed by driving force F_{t+1} , which is the sum of the direction and magnitude of two driving forces: FD_{t+1} (force that drives price down of day $t + 1$) and FU_{t+1} (force that drives price up of day $t + 1$) as defined below.

$$CP_{t+1} = \begin{cases} P_t < P_{t+1} (\text{increasing}) \\ P_t > P_{t+1} (\text{decreasing}) \\ P_t = P_{t+1} (\text{nochange}) \end{cases} \begin{cases} \text{if } FD_{t+1} < FU_{t+1} \\ \text{if } FU_{t+1} < FD_{t+1} \text{ or } \\ \text{if } FU_{t+1} = FD_{t+1} \end{cases} \begin{cases} \text{if } F_{t+1} > 0 \\ \text{if } F_{t+1} < 0 \\ \text{if } F_{t+1} = 0 \end{cases} \quad (1)$$

where CP_{t+1} is price increase or price decrease, F_{t+1} is driving force of the price of day $t + 1$, and

FD_{t+1}, FU_{t+1} are the forces that drive the price of day $t + 1$ down and up, respectively.

As a classification problem, price change of day $t + 1$ can be defined as binary outcome as follows.

$$Y_{t+1} = \begin{cases} 1, \text{ if } P_t < P_{t+1} \\ 0, \text{ if } P_t \geq P_{t+1} \end{cases} \quad (2)$$

The predicted outcome Y_{t+1} is a nonlinear function of the set of input features X_t of day t

$$Y_{t+1} = f(X_t) \quad (3)$$

where f is a nonlinear function that maps a set of input features X_t at day t to binary outcome values $\{0, 1\}$ of day $t + 1$ (Yun et al., 2021).

3.2. The outline of the proposed methodology

The proposed methodology is outlined as follows and depicted in Fig. 1.

Module1: Data preparation and cleaning.

- Historical stock prices ('High', 'Low', 'Open', 'Close', 'Volume') and selected price-derived technical indicators are combined with VIX, PCR, FGX for dataset 2, dataset 3, and dataset 4 as shown in Table2 through 4 in Section 3.6.2.

Table 2

Price-derived technical indicators in dataset2.

Technical indicators	Definition
Double Exponential Moving Average (DEMA)	$2EMA_{26} - EMA(EMA_{12})$
Balance of Power (BOP)	$\frac{C_t - O_t}{H_t - L_t}$
Average Price (AVP)	$\frac{O_t + C_t + L_t + H_t}{4}$
Typical Price (TYP)	$\frac{C_t + L_t + H_t}{3}$
Weighted Close Price (WCP)	$\frac{2C_t + L_t + H_t}{4}$
Median Price (MDP)	$\frac{L_t + H_t}{2}$

Note: C_t : Close price at day t . O_t : Open price at day t . H_t : High price at day t . L_t : Low price at day t . EMA_t : 14-day exponential moving average at day $t = C_t \times \frac{2}{14 + 1} + EMA_{t-1} \times \frac{12}{14 + 1}$.

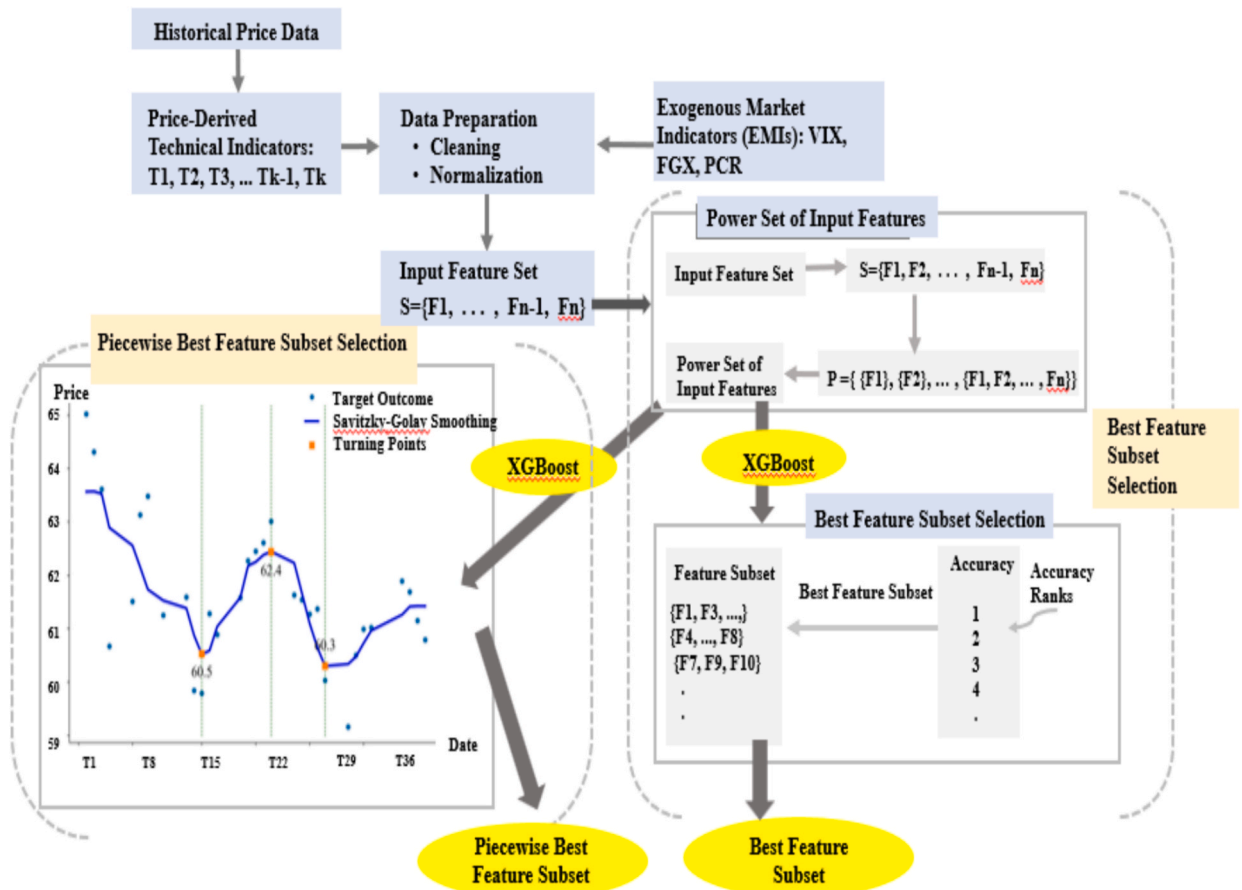


Fig. 1. BFS and PBFS selection process.

Table 3

Price-derived technical indicators in dataset3.

Technical indicators	Definition
Momentum (MOM)	$C_t - C_{t-14} \times 100$
Relative strength index (RSI)	$100 - \frac{100}{1 + (\sum_{i=0}^{n-1} Up_{t-1}/14) / (\sum_{i=0}^{n-1} Dw_{t-1}/14)}$
Commodity channel index (CCI)	$\frac{M - m}{d \times 0.015} \times 100$
Stochastic %K (STK)	$\frac{C_t - LL_{t-14}}{HH_{t-14} - LL_{t-14}} \times 100$
Larry William's %R (LWR)	$\frac{H_{14} - C_t}{H_{14} - L_{14}} \times 100$
Moving average convergence divergence (MACD)	$\frac{EMA_{12} - EMA_{26}}{H_t - C_{t-1}} \times 100$
Accumulation/Distribution oscillator (ADR)	$\frac{H_t - L_t}{H_t - L_t} \times 100$

Note: C_t : the closing price at day t. Up_t : the upward price change at day t. Dw_t : the downward price change at day t. M_t : typical price at day t = $(H_t + L_t + C_t)/3$. m_t : 14-day moving average = $(\sum_{i=1}^{14} M_{t-i+1})/14$. d_t : 14 days' mean deviation = $(\sum_{i=1}^{14} |M_{t-i+1} - m_t|)/14$. LL_t : the lowest low price in the last t days. HH_t : the highest high price in the last t days. Dw_t : the downward price change at day t. H_t : the highest price at day t. L_t : the lowest price at day t. EMA_t : 14-day exponential moving average at day t = $C_t \times 2/14 + EMA_{t-1} \times (1 - 2/(14 + 1))$.

Table 4

Price-derived technical indicators in dataset4.

Technical indicators (Abbreviation)	Definition
Balance of Power (BOP)	$\frac{C_t - O_t}{H_t - L_t} \times 100$
Relative Strength Index (RSI)	$100 - \frac{100}{1 + (\sum_{i=0}^{n-1} Up_{t-1}/14) / (\sum_{i=0}^{n-1} Dw_{t-1}/14)}$
Stochastic Relative Strength Index High (STOK)	$\frac{RSI_{14} - LLRSI_{14}}{HRSI_{14} \times LLRSI_{14}}$
Hillbert Transform –Inphase (INPHASE)	$\frac{EMA_{12} - EMA_{26}}{H_t - L_t} \times 100$
Stochastic Fast High (STOCHF)	$\frac{ (C_t - L_t) - (H_t - C_t) }{H_t - L_t}$
Chaikin A/D Oscillator (DOSC)	$\frac{\max[(H_t - L_t), H_t - C_{t-1} , L_t - C_{t-1}]}{100}$
True Range (TRANGE)	$100 - \frac{100}{1 + PMF_{14}/NMF_{14}}$
Money Flow Index (MFI)	

Note: C_t : the closing price at day t. O_t : Open price at day t. H_t : High price at day t. L_t : Low price at day t. Up_t : the upward price change at day t. Dw_t : the downward price change at day t. MA_t : Moving Average of t days. LL_t : the lowest low price in the last t days. HH_t : the highest high price in the last t days. EMA_t : 14-day exponential moving average at day t = $C_t \times 2/14 + EMA_{t-1} \times (1 - 2/(14 + 1))$. PMF_t : Positive Money Flow of t days. NMF_t : Negative Money Flow of t days.

- Creating binary outcome variable according to the definition in Formula (2).
- Removal of missing observations and 'Close' price from the input features.
- Normalization of the input values using min-max Formula (15)

Module2: BFS selection

- Power set of input features are constructed for each dataset.
- XGB evaluates each non-empty feature subset of power set for prediction accuracy.
- The BFS is selected according to the accuracy ranks of all feature subsets

Module3: PBFS selection

- Savitzky-Golay smoothing filter is used to reduce noises and outliers and construct piecewise time segments for target periods
- XGB evaluates each non-empty feature subset of power set for prediction accuracy of each time segment
- The PBFS is selected according to the accuracy ranks of each time segment.

3.3. BFS selection

Algorithm 1: The BFS selection.

```

Input F: historical stock price and price-derived technical indicators
E: exogenous market sentiment indicators
S = F ∪ E = {f1, f2, ..., fn}
Output B: BFS of S
1. Initialize BFS = { }
2. Construct the power set of S: P = { { }, {f1}, {f2}, ..., {f1, f2, ..., fn} }
3. Remove the empty subset of P
4. For k = 1, 2, ..., 2n - 1, do
    Calculate the fitness value (accuracy) of each subset of P using XGB classifier
    Update the BFS based on the fitness value (accuracy)
End

```

Let S be the set of n input features f_i , where $i = 1, 2, 3, \dots, n$. The set S is a union set of k price-derived technical indicators and n-k exogenous market indicators. Then, S is defined as

$$S = \{f_1, f_2, \dots, f_n\} \quad (4)$$

To select the BFS out of the set of n input features, the power set of S without the nonempty subset is constructed as

$$P = \{\{f_1\}, \{f_2\}, \dots, \{f_1, f_2, \dots, f_n\}\} \quad (5)$$

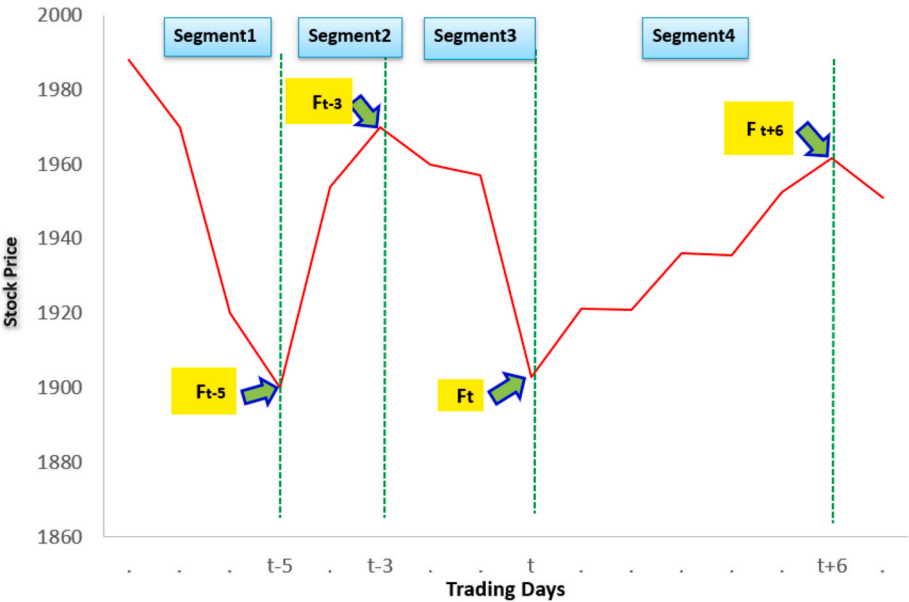


Fig. 2. Piecewise segmentation of price behaviour.

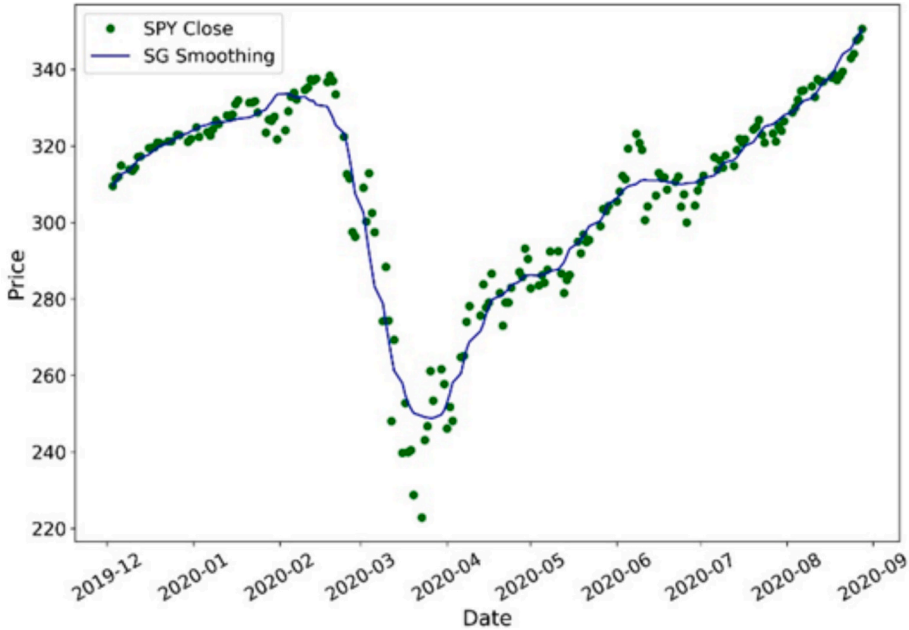


Fig. 3. Noise reduction by SG smoothing filter.

where the number of elements in P is $2^n - 1$. Each subset of P , which is a non-empty element of the power set of S , is evaluated by XGB classifier for the best prediction accuracy. The subset with the best prediction accuracy is selected as the BFS of the input feature set S as described in Fig. 1.

3.4. PBFS selection

Algorithm 2: The PBFS selection.

Input F: historical stock price and price-derived technical indicators
 E: exogenous market sentiment indicators
 $S = F \cup E = \{f_1, f_2, \dots, f_n\}$
Output B: BFS of S

1. **Construct** the piecewise time segments of target period using Savitzky-Golay smoothing
2. **Initialize** PBFS = { }
3. **Construct** the power set of S : $P = \{\{\}, \{f_1\}, \{f_2\}, \dots, \{f_1, f_2, \dots, f_n\}\}$
4. **Remove** the empty subset of P
5. **For** each time segment ($i = 1, 2, \dots, t$), **do**
 For $k = 1, 2, \dots, 2^n - 1$, **do**
 Calculate the fitness value (accuracy) of each subset of P using XGB classifier
 Update the PBFS based on the fitness value (accuracy)

End

3.4.1. Piecewise segmentation of price behaviour

Stock price is affected by the complex dynamics of various factors such as political, economic, social news, and public sentiment (Yun et al., 2021). Under the assumption that the price of day $t + 1$ is changed by driving force F_{t+1} that is the sum of FD_{t+1} (force that drives price down at day $t + 1$) and FU_{t+1} (force that drives price up at day $t + 1$), the price P_{t+1} maintains the same behaviour state of increase (positive) or decrease (negative) until a new driving force with greater magnitude of opposite direction (sign) makes a reverse state. The change of price behaviour occurs at turning points.

Fig. 2 shows turning points F_{t-5} , F_{t-3} , F_t , and F_{t+6} , where the state of price behaviour changes. Each time segment from one turning point to the next turning point keeps consistent price behaviour: increase or decrease. It implies that the relationship between input features and the outcome remains consistent for each piecewise time segment, once it is changed at a previous turning point. Hence, piecewise price behaviour analysis between two turning points is desirable to capture the purity of the cause and effect in the period. Piecewise effect analysis is a piecewise regression in which the relationship between outcome variable and input features abruptly changes at certain points that are called break-points or thresholds (Yun et al., 2023; Lu & Chang, 2023).

3.4.2. Savitzky-Golay smoothing filter

To construct the piecewise time segment of price function as the preliminary step of piecewise effect analysis, Savitzky-Golay (SG) smoothing filter is used. SG filtering can reduce unnecessarily frequent noises of price function. Volatile and constantly changing daily stock price is discrete and makes many ups and downs. Trading at every turning point misspends unnecessary higher transaction costs and loses opportunity of greater return (Chiang, Enke, Wu, & Wang, 2016; Yun et al., 2023). SG smoothing filter is used to reduce noises of discrete price data while keeping crucial information about the shape and turning points of price function as shown in Fig. 3.

SG smoothing filter was proposed by Savitzky and Golay (Savitzky & Golay, 1964) to smooth a sequence of consecutive values that are called a spectrum. For the given spectrum, a weighted moving average, called convolution, is calculated as a polynomial with higher moments within the points. The equation of the least-squares convolution is given as follows:

$$Y_j^* = \sum_{i=-m}^m C_i Y_{j+i} / N \quad (6)$$

where Y is the given data points, Y_j^* is the filtered value, C_i is the

coefficient of the filter, m is any positive integer, and N is the number of convoluting values that is equal to $2m + 1$ (Chen et al., 2004).

3.4.3. PBFS selection

Once the turning points of stock price function are identified for target period by using the SG smoothing filter, then piecewise time segmentations are constructed to reflect the consistent price behavior of each time segment identified by the turning points. As described in 3.3.1., the relationship between input features and the outcome stock price remains consistent for each piecewise time segment. Hence, price function's behavior also stays the same: increase or decrease until a new driving force with greater magnitude of opposite direction changes the current price behavior. To select the BFS for each time segment, all non-empty elements in the power set of all input features, that is defined as P in Formula (5), are evaluated by XGB for each time segment.

The subset with the best prediction accuracy is selected as the PBFS as described in Fig. 1 in Section 3.2.

3.5. XGB classifier

XGB classifier is adopted as the learning algorithm to predict the movement direction of daily stock price. Extreme gradient boosting, also known as XGBoost or XGB, is developed by Chen & Guestrin in 2016 (Chen & Guestrin, 2016). It is one of the most popular tree-structured ML algorithms because of its fast-learning process and outstanding accuracy. The exceptional performance of XGB has been proved in many stock price direction prediction studies (Chung & Shin, 2018; Yun et al., 2021). The strength of XGB is its scalability integrated with parallel and distributed computing (Chen & Guestrin, 2016). The scalability of XGB is easily capable of handling a growing amount of work process. Its parallel and distributed computing method by using block structure reduces memory usage for large datasets. Hence, XGB simply maximizes the processing speed and performance by making the most of every available computing resource (Yun et al., 2021).

The good performance of XGB has been proved in many recent stock price prediction studies (Chung & Shin, 2018; Chung & Shin, 2020; Yun et al., 2021; Park et al., 2021). In particular, a previous study empirically showed that the prediction performance of XGB substantially outperforms other competitive ML algorithms: Random Forest, Extra Trees, Decision Tree, Support Vector Machine with RBF kernel, and K-Nearest Neighbour classifies in stock price direction prediction (Yun et al., 2021). Furthermore, combined with genetic algorithm, their XGB classifier's prediction accuracy and learning time overwhelms the accuracy and process time of various LSTM and Bidirectional LSTM models in stock price direction prediction (Yun et al., 2021). Additionally, tree-structured ML classifiers such as XGB, Random Forest, Extra Trees, Decision Tree provides superior interpretability compared to DL algorithms because they keep input features without any transformation (Linardatos et al., 2020; Yun et al., 2023). The interpretability of learning outcome is crucial in the practice of stock market investment as well as stock price prediction. For this reason, XGB algorithm is adopted as the prediction algorithm in this study.

Once SG filter produces a smooth curve of noisy price data and turning points for local extrema, then piecewise time segment is constructed for the period to be tested. XGB is trained on an input training dataset and predicts stock price movement direction for each piecewise segment of the test data. For each piecewise time segment, XGB ranks all non-empty subsets of input features by the corresponding prediction accuracy and returns the feature importance of best performing feature subset. As the result, piecewise effect of exogenous market indicators is measured by the contribution to the prediction accuracy in relation to other features in a dataset.

3.6. Input features

Historical stock prices and technical indicators including price-

derived technical indicators and exogenous market indicators are time-series data. Conventional statistical time-series analysis underlines the characteristics of time-series data: stationarity, trend, and seasonality. Particularly, stationarity assumption is important in classic time series methods. However, financial market price data generally do not bear these characteristics. To produce a reliable analysis using statistical time-series methods, transformation of time-series data is sometimes applied, and there are several transformation methods to treat data that do not meet underlying assumptions (Salles et al., 2019). Nevertheless, it has been shown by numerous studies that the prediction accuracy of statistical time series analysis such as such as autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), and autoregressive conditional heteroscedastic model (ARCH) underperforms ML and DL methods (Chung & Shin, 2018). ML and DL-based stock prediction studies relieve the inconvenience of satisfying underlying statistical assumptions for learning and prediction. Instead, engineering the optimal feature space or constructing an effective prediction algorithm are more important in ML and DL methods (Yun et al., 2021). Especially, appropriate feature engineering: feature generation, feature transformation, feature extraction, and feature selection is a proven way to improve prediction accuracy and interpretability in various ML models. Hence, we use the following input features in this study.

3.6.1. Exogenous market sentiment indicators

The main goal of this study is to analyse the effect of exogenous market sentiment indicators in stock price prediction. To do this, the importance and performance of three exogenous market sentiment indicators: VIX, FGX, and PCR are measured and compared with selected price-derived technical indicators in different datasets.

- VIX (implied volatility)

It is created by the Chicago Board Options Exchange (CBOE) to measure the expected 30-day volatility in the U.S. stock market. It is defined as

$$VIX = \sqrt{\frac{2e^{nr}}{n} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right)} \quad (7)$$

where n is the number of mean days in a month, r is the risk-free rate, F is the 30-day forward price on the S&P 500, and $P(K)$ and $C(K)$ are put and call price with strike price K and 30 days maturity.

The VIX in the range between 0 and 15 means very low volatility and optimism in the market, 15 to 25 means moderate volatility for normal market environment, 25 to 30 means some market turbulence with significant volatility, and over 30 means high volatility with some extreme swings in the market.

(VIX Explained, 2023).

- Fear and Greed Index (FGX)

Developed by CNNMoney and is used to gauge stock market movements and if stock price is a fair price or not. It is defined as

$$FGX = \frac{\sum_{x=1}^7 I_x}{7} \quad (8)$$

where I_x , $x = 1, 2, \dots, 7$ and $0 \leq I_x \leq 100$.

Each I_x represents stock price momentum, stock price strength, stock price breadth, put and call options, junk bond demand, market volatility, and safe haven demand, respectively (Fear & Greed Index, 2024).

- o Stock Price Momentum: A measure of the S&P 500 versus its 125-day moving average (MA).

- o Stock Price Strength: The number of stocks hitting 52-week highs versus those hitting 52-week lows on the New York Stock Exchange (NYSE).
- o Stock Price Breadth: Analyzing the trading volumes in rising stocks against declining stocks.
- o Put and Call Options: The extent to which put options lag behind call options, signifying greed, or surpassing them, indicating fear.
- o Junk Bond Demand: Measures the spread between yields on investment-grade bonds and junk bonds.
- o Market Volatility: CNN measures the CBOE's Volatility Index (VIX) based on a 50-day MA.
- o Safe Haven Demand: The difference in returns for stocks versus treasuries.

The interpretation of FGX is as follows. FGX score between 0 and 24 means extreme fear, 25 to 44 means fear, 45 to 55 means neutral, 56 to 75 means greed, and 76 to 100 means extreme greed.

- Put-call ratio (PCR)

It is the ratio of number of put-option trades over number of call-option trades. It is defined as

$$PCR = \frac{\text{Putoptiontradingvolume}}{\text{Calloptiontradingvolume}} \quad (9)$$

The PCR value of 1 literally means the number of put buying is the same as the number of call buying, but it is natural to buy call options more than put options. Hence, 0.7 value is considered as a basis for evaluation the overall sentiment of a market as follows (Put-Call Ratio, 2024).

- o PCR > 0.7: More put options purchased than call options. Market anticipates a bearish outlook.
- o PCR < 0.7: Less put options purchased than call options. Market anticipates a bullish market.

3.6.2. Price-derived technical indicators and datasets

To measure the effect of EMIs in interactions with other price-derived technical indicators in a dataset, four sets of price-derived technical indicators are selected. Dataset1 is composed of historical stock prices and three exogenous market indicators: VIX, FGX, PCR. Dataset2 through dataset4 are adopted from previous studies for comparison purposes to show the importance and contribution of three EMIs over datasets.

Dataset2 has Double Exponential Moving Average (Dema), Balance of Power (BOP), Low price, Average price (AVP), Typical price (TYP), Weighted Close price (WCP), Median price (MDP), and VIX, FGX, and PCR. The technical indicators of Dataset2 are adopted from previous stock price prediction studies (Yun et al., 2023). They selected these features as optimal features through the competition of five ML regression algorithms. The definitions of selected technical indicators are described in Table 2. Dema is a momentum indicator and the combination of two exponential moving average. It measures price's trends reversals. BOP indicates the direction of a trend measuring the strength of buyers against sellers by assessing the ability of each side to drive prices to an extreme level (Yun et al., 2021). AVP, TYP, WCP, MDP are transformed price of base historical prices. They provide another aspect of day prices without removing noise and outliers.

Dataset3 also has 7 price-derived technical indicators: Momentum of 14 trading days (MOM). Relative strength index (RSI), Commodity channel index (CCI), Stochastic %K (STK), Larry William's %R (LWR), Moving average convergence divergence (MACD), Accumulation/ Distribution oscillator (ADR). They are commonly used technical indicators in previous stock price prediction studies (Basak et al., 2019; Chung & Shin, 2018; Chung & Shin 2020; Yun et al., 2021). The definition and description of each price-derived technical indicators are shown in

Table 5

Contingency table of binary classification.

		Actual Class	
		P	N
Predicted Class	P	TP	FP
	N	FN	TN

Table 3. MOM is the difference of the current closing price and the closing price of 14 trading days ago. It reflects the trend (strength or weakness) of the last 14 days' stock price. RSI determines whether the stock is overbought or oversold using the range from 0 to 100. If a stock is overbought, RSI gets closer to 100, and the stock is oversold, then RSI gets closer to 0. CCI measures the difference between the current price and the historical average price. When $CCI > 0$, it indicates the current stock price is higher than the historical average price. Otherwise, the stock price is lower than the historical average.

STK measures the position (between 0 and 100) of a stock's closing price as a stochastic oscillator. If the value is 80 or above, it indicates the overbought range, and if the value is 20 or lower, then it is considered oversold. LWR measures the level of a market's closing price with the highest price for 14 days using the range between 0 and 100. When the value > 20 , it is considered as a sell signal and when its value < 80 it is considered as a buy signal. MACD compares two moving averages of prices by subtracting the 26-day Exponential Moving Average (EMA) from the 12-day EMA. It represents the signal for overbought and oversold. When it crosses above (to buy) or below (to sell) its signal line. ADR uses volume and price to imply a rising price trend, or a falling price trend based on its slope of the line (Yun et al., 2021).

Dataset4 is borrowed from the previous study (Yun et al., 2021). They selected 33 optimal features out of 67 price-derived technical indicators and 4 historical stock prices (High, Low, Open, and Volume) for stock price direction prediction through a hybrid GA-XGB algorithm. Out of the 33 optimal features they selected, the most important 10 features only are borrowed for this study and shown in Table 4. BOP and RSI of dataset4 are the same technical indicators of dataset3, and STOK is the high Stochastic RSI whose value is above 80 and indicates overbought of a stock. It can be considered as a signal for a price pullback or imminent reversal. INPHASE is the difference between the past 12 days' EMA and the past 26 days' EMA. It implies the price trend change of between the target trading days. STOCHF measures the momentum of a price that shows the strength of price trends. DOSC is the magnitude of the deviations of daily stock price change. It measures the strength of the daily stock price change as the ratio. TRANGE is the volatility of a daily stock price. The larger its value is, the greater the price fluctuates. MFI is a measure of buying and selling pressure that is measured the price and volume. It is positive when the typical price rises (buying pressure) and negative when the price declines (selling pressure) (Yun et al., 2021).

3.7. Evaluation metrics

For the performance evaluation of input features for each piecewise time segment, the following conventional metrics are used.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (11)$$

Table 6

Hyperparameters of XGB classifier.

max_depth	learning_rate	n_estimators	booster	gamma	min_child_weight	subsample	nthread	colsample
3	0.1	100	gbtree	0	1	1	none	1

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$AUC = \int_0^1 \frac{TP}{TP + FN} d \frac{FP}{FP + TN} \quad (14)$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives.

The contingency table is used to measure the performance of binary classification as shown in Table 5. Accuracy is the percentage of correct predictions out of total number of predictions, whether positive or negative, as shown in Formula (10). A perfect classification would be 100 % with zero false positives and zero false negatives. When the dataset is balanced, accuracy is a fair measure of overall correctness of a model. However, accuracy might be misleading when the dataset is severely imbalanced. F1 is the harmonic mean of precision and recall as shown in Formula (11). F1 considers both false positives and false negatives important, while accuracy considers true positives and true negatives important. Hence, F1 works well with imbalanced class data. Precision is the rate of correct positive predictions out of the total number of positive predictions, whether correct or incorrect, as shown in Formula (12). If the false positive is zero, then precision is perfect 1. Recall is the rate of correctly predicted positives out of all actual positives as shown in Formula (13). Precision and recall work well with imbalanced class data. AUC is the area under the Receiver Operating Characteristic (ROC) curve as described in Formula (14). It represents the probability that the model will rank the positive than the negative. AUC of 1 means that the model correctly ranks a positive example higher than a negative example with 100 %. AUC is useful for comparison different balanced models.

4. Experimental results and discussion

4.1. Experiment setting and data description

Python of version 3.9 is used on Windows 11 for this study. To measure the importance and effect of EMIs in the collective behaviour of input features, 4 different datasets of SPDR S&P 500 ETF Trust (Ticker: SPY) stock prices were constructed by combining EMIs and respective sets of price-derived technical indicators as described in Section 3.5.2. Dataset1 through dataset4 used SPY historical stock prices for the period of 1/2/2015 through 6/30/2020 (total 1382 trading days). Each dataset is divided into the first 1237 days (2015/1/2 ~ 2019/11/30) and the last 145 days (2019/12/1 ~ 2020/6/30) in order. The first part is used as training data (the first 1052 days) and validation data (the last 185 days), and the holdout 145 trading days (2019/12/1 ~ 2020/6/30) are used for the final test data. The XGB classifier uses the following default hyperparameters without tuning as shown in Table 6.

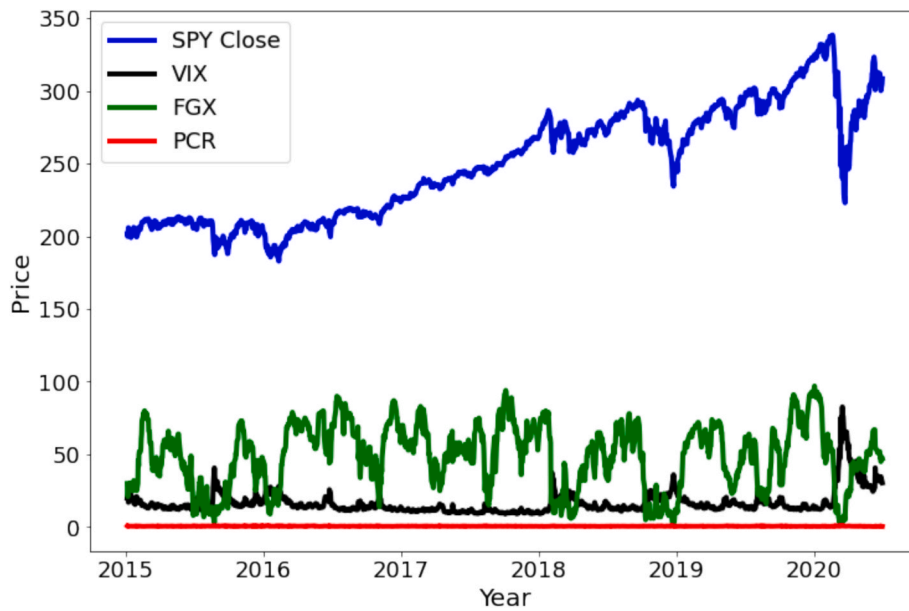
The built-in hyperparameters are used for the XGB without hyperparameters tuning. This study focuses on the effect and contribution of three EMIs in the best feature set by measuring model's prediction accuracy and the importance of three EMIs. The importance and contribution of the EMIs are greatly related to input features and the model's prediction algorithm with feature importance scores. In this case, feature importance is more robust without the optimization of the hyperparameters.

In addition, ML algorithms such as XGB, Random Forest, Extra Trees,

Table 7

Summary statistics of dataset1.

	Open	High	Low	Close	Adj Close	Volume	VIX	FGX	PCR
count	1383	1383	1383	1383	1383	1,383	1383	1383	1383
mean	249.93	251.2	248.54	249.96	225.42	96,734,089	16.73	49.28	0.65
std	38.24	38.44	38.02	38.21	40.99	52,834,104	8.15	22	0.1
min	182.34	184.1	181.02	182.86	158.81	20,270,000	9.14	2	0.37
25 %	211.06	211.77	210.18	211	182.11	62,110,000	12.41	33	0.58
50 %	247.96	249.71	246.97	248.07	224.92	82,182,300	14.42	52	0.63
75 %	281.22	283.02	279.99	281.38	258.75	114,395,850	18.06	66	0.7
max	337.79	339.08	337.48	338.34	317.71	507,244,300	82.69	97	1.28

**Fig. 4.** Historical SPY Price, VIX, Fear-Greed, Put-Call Indices (2015–2020).**Table 8**

Correlation coefficients between features.

	Close	Outcome
VIX	0.039	0.090
FGX	0.132	0.171
PCR	−0.336	−0.421

Decision Tree model are already quite robust, and they can provide competitive performance without hyperparameters tuning. To compare different datasets using various ML algorithms, hyperparameters tuning is not used for fair comparison purpose. For the objective of this study, the prediction accuracy of datasets and the feature importance scores of three EMIs are explored and compared over different datasets to validate the generality of the EMI's importance and contribution. To compare different datasets with three EMIs, using default hyperparameters of XGB is desired for fair comparison purposes. In particular, the importance and contribution of the three EMIs are greatly related to input features of datasets. In this case, feature importance is more robust with default hyperparameters.

The descriptive statistics of dataset1, which are comprised of basic historical prices and three EMIs, are shown in Table 7. PCR value ranges from 0.37 to 1.28, which is notably different from the ranges of VIX and FGX.

Fig. 4 is the graph of the daily 'Close' price of SPY with three exogenous market indicators for the training and testing period. As shown in the Fig. 4, VIX and FGX make sudden sharp changes for the outbreak of Covid-19 pandemic in the first half of 2020, but the change of PCR graph

is indiscernible in the plot because it has different measurement range as shown in the Table 7 above.

4.2. Data processing

Once dataset 1 through dataset4 are constructed by combining selected price-derived technical indicators with VIX, PCR, FGX, then 'Close' price of each dataset is replaced by the binary outcome variable by which the movement direction of stock price is represented. The value 1 of the outcome variable means price increases from day $t-1$ to t and 0 means the price does not increase from day $t-1$ to t . After the removal of 'Close' price and missing valued observations, all datasets were normalized by the min-max formula as shown in Formula 15.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (15)$$

To find the BFS, the power set of each dataset is constructed as described in Section. Next, the XGB evaluates each nonempty subset of the power set for prediction accuracy. The BFS is selected according to the accuracy ranks of all feature subsets. For the piecewise best feature subset selection, Savitzky-Golay smoothing filter is used to reduce noises and outliers and construct piecewise time segments for target periods. The XGB evaluates each feature subset of power set for prediction accuracy of each time segment. The PBFS of each time segment is selected according to the accuracy ranks of all feature subsets as shown in Fig. 1 and described in Section 3.4.

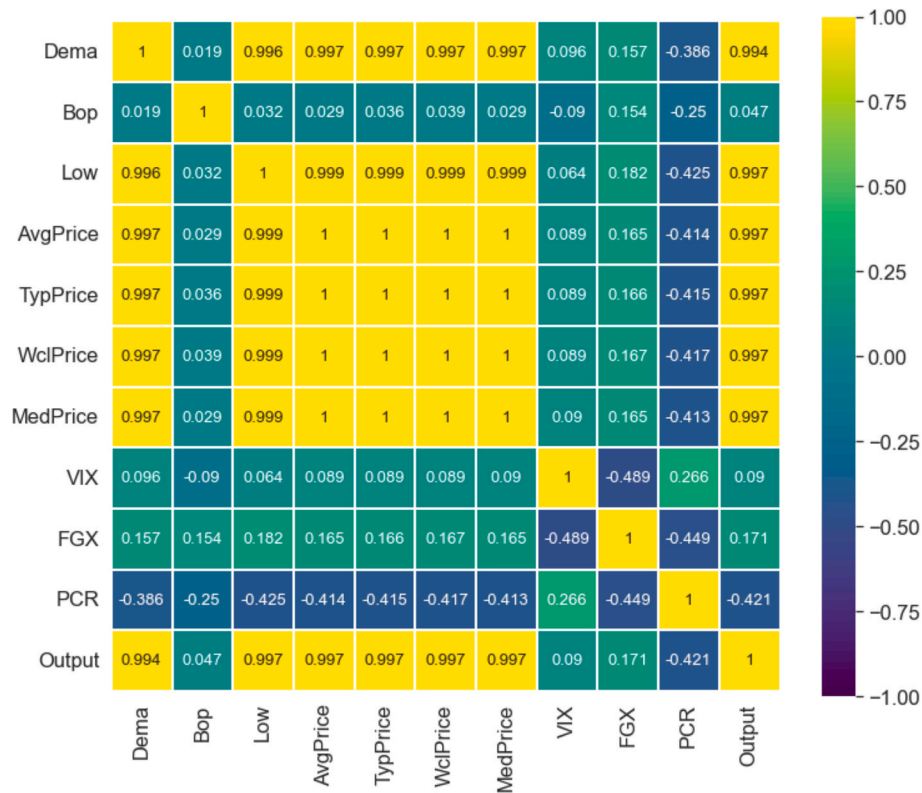


Fig. 5. Correlation coefficient map of features in dataset 2.



Fig. 6. Correlation coefficient map of features in dataset 3.

Table 9

Performance comparison over test datasets (12/1/2019–6/30/2020).

	Dataset1		Dataset2		Dataset3		Dataset4	
	No-EMIs	With-EMIs	No-EMIs	With-EMIs	No-EMIs	With-EMIs	No-EMIs	With-EMIs
Accuracy	54.67 %	61.19 %	78.47 %	81.87 %	84.42 %	86.97 %	95.18 %	95.47 %
Precision	63.86 %	73.15 %	81.25 %	82.87 %	94.15 %	93.48 %	96.10 %	95.67 %
Recall	51.46 %	52.91 %	82.04 %	86.89 %	78.16 %	83.50 %	95.63 %	96.60 %
AUC score	56.99 %	61.41 %	77.75 %	80.86 %	85.68 %	87.67 %	95.09 %	95.24 %
F1 Score	55.32 %	62.85 %	81.64 %	84.83 %	85.41 %	88.21 %	95.86 %	96.14 %
Best Features	Open	PCR	Bop	Bop	Stok	Stok	Stok	Stok
	Low	Open	WclPrice	PCR	Cci	Cci	Bop	Bop
	Volume	High	Dema	Low	Rsi	Willr	Stod	Stod
			Low	VIX	Willr	PCR	Slowd	PCR
			TypPrice	Dema	Mom	Macd	Slowk	
				FGX		Mom	Stochfh	
						VIX	Trange	

4.3. Correlation check

Pearson correlation coefficients between ‘Close’ price and three EMIs, and ‘Outcome’ variable and three EMIs are measured as the preliminary data exploration, and they are shown in Table 8.

Pairwise Pearson correlation coefficients are used to measure the linear relationships between input technical indicators and the target stock price. The direction and strength of correlation between two variables are interpreted by the sign and the magnitude of the correlation coefficient value that ranges from -1 that represents perfect negative linear relationship and 1 that represents perfect positive linear relationship. The correlation coefficient around 0 has no or very weak linear relationship between two variables. For interpretation, a positive correlation coefficient means that as a variable increases or decreases, the other variable also increases or decreases in the same direction. Negative correlation coefficient implies that as a variable increases or decreases, the other variable moves in the opposite direction. The stock price is widely known for its complex nonlinearity (Chen & Hao, 2017; Chung & Shin, 2018; Basak, Kar, Saha, Khaidem, & Dey, 2019; Henrique, Sobreiro, & Kimura, 2019). The nonlinearity of stock price data implies that the relationship between a stock price and the timeline does not exclude the potential linear relationship between input features and

the stock price as the outcome. Hence, a stock price might have a linear relationship with input features. Checking the correlation between input variables and the outcome stock price has been used to identify important input features, price behaviour, and potential multicollinearity among input features in many stock price prediction studies (Thakkar, Patel, & Shah, 2021; Li and Pan, 2022). For this reason, this study also uses Pearson correlation coefficients to check the pairwise linear relationship among the input features and outcome stock price as a preliminary data analysis. As the result, three EMIs: ‘VIX,’ ‘FGX,’ and ‘PCR’ have no significant linear relationships with the same day ‘Close’ price nor with the outcome variable that is the following day’s ‘Close’ price.

The linear relationships between each pair of input feature and the outcome variable in dataset2 and dataset3 are shown in Fig. 5 and Fig. 6.

VIX and FGX have very weak linear relationships with all price-derived technical indicators and the outcome variable in dataset2. Meanwhile, PCR has moderate linear relationships with price-derived technical indicators: Dema, Low, AvgPrice, TypPrice, WclPrice, MedPrice, and the outcome variable. In dataset3, there are slightly higher linear relationships between pairs of EMIs and price derived technical indicators than in dataset2, but only FGX has a strong correlation (0.836) with RSI.

The correlation coefficient map of dataset4 is measured and shown in

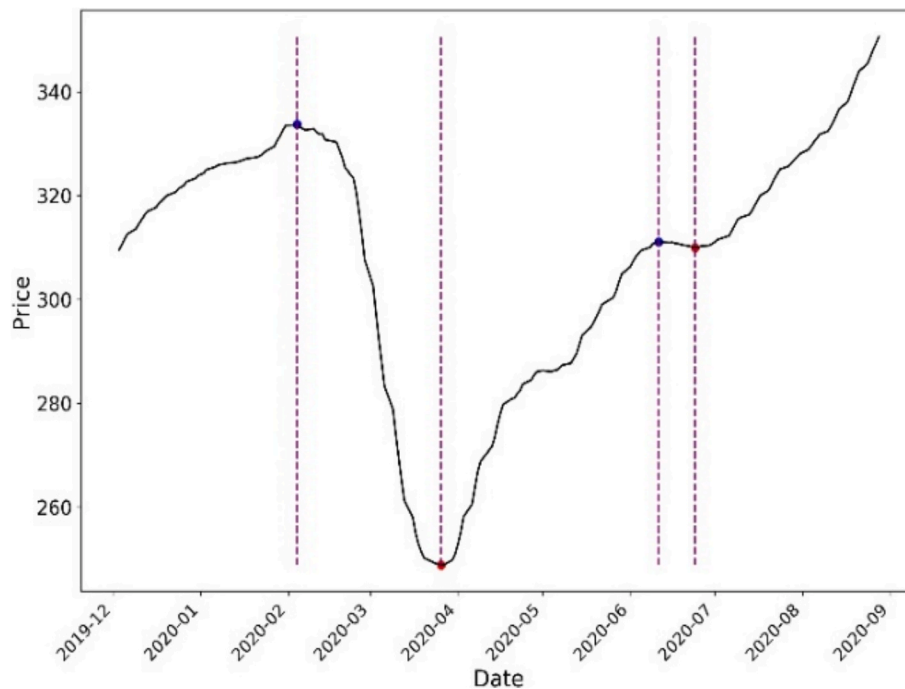


Fig. 7. Piecewise time segmentation by SG smoothing.

Table 10

Piecewise performance of XGB and PBFS on dataset3 and dataset4.

Data	Methods	Part1 12/1/2019–2/4/2020		Part2 2/4/2020–3/26/2020		Part3 3/26/2020–6/10/2020		Part4 6/10/2020–6/30/2020		All Parts 12/1/2019–6/30/2020	
		XGB	PBFS	XGB	PBFS	XGB	PBFS	XGB	PBFS	XGB	PBFS
Dataset3	Accuracy	0.8636	0.9091	0.8649	0.9730	0.8679	0.9434	0.8667	0.9333	0.8562	0.9041
	Precision	0.8929	0.9286	0.8000	1.0000	1.0000	1.0000	0.8889	1.0000	0.9091	0.9487
	Recall	0.8929	0.9286	0.9412	0.9412	0.7812	0.9062	0.8889	0.8889	0.8333	0.8810
	F1	0.8929	0.9286	0.8649	0.9697	0.8772	0.9508	0.8889	0.9412	0.8696	0.9136
	AUC	0.8527	0.9018	0.8706	0.9706	0.8906	0.9531	0.8611	0.9444	0.8602	0.9082
Dataset4	Accuracy	0.9545	0.9773	0.9444	0.9722	0.9808	1.0000	1.0000	1.0000	0.9315	0.9521
	Precision	0.9643	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9405	0.9639
	Recall	0.9643	0.9643	0.8750	0.9375	0.9677	1.0000	1.0000	1.0000	0.9405	0.9524
	F1	0.9643	0.9818	0.9333	0.9677	0.9836	1.0000	1.0000	1.0000	0.9405	0.9581
	AUC	0.9509	0.9821	0.9375	0.9688	0.9839	1.0000	1.0000	1.0000	0.9299	0.9520

the [Appendix B](#). Dataset4 also has no significant linear relationship between EMIs and other input features including the outcome variable. Only the pair of VIX and Trange has a strong correlation of 0.843.

The result of the preliminary correlation check between input features and the outcome implies that the stock price prediction accuracy does not dependent on linear relationship because stock price data are nonlinear ([Chung & Shin, 2018](#); [Yun et al., 2021](#)). Instead, it rather depends on the collective behaviour and complex interactions among input features.

4.4. Prediction accuracy comparison of datasets

The BFS's prediction accuracy of 1-day stock price movement direction for the testing period of each dataset was measured in two ways: with EMIs and without EMIs. The comparison result is shown in [Table 9](#). The prediction accuracy with EMIs is higher than that without EMIs of each dataset. Especially, the lower the prediction accuracy of dataset is, the greater the accuracy increment EMIs add. For example, Dataset1 has prediction accuracy of 54.57 % without EMIs but 61.19 % with 'PCR'. The prediction accuracy of Dataset2 without EMIs is 78.47 % but it is increased to 81.87 % with the inclusion of all three EMIs 'PCR,' 'VIX,' 'FGX.' In Dataset3 and Dataset4, prediction accuracy increment by including EMIs is not so large as in Dataset1 and Dataset2. Dataset3's prediction accuracy is changed to 86.97 % from 84.42 %, and Dataset4's prediction accuracy is changed to 95.47 % from 95.18 % with the inclusion of EMIs. In common, the inclusion of EMIs increases prediction accuracy. Among the 3 EMIs, 'PCR' is most frequently selected in the BFS of each dataset as shown in [Table 9](#).

4.5. Piecewise effect analysis

As the preliminary step of the PBFS selection method, Savitzky-Golay (SG) Smoothing was rendered, and the result was depicted with the Close price in [Fig. 7](#). Testing period between 12/1/2019 and 6/30/2020 (Total 146 trading days) was divided into 4 piecewise time segments as the result of the SG smoothing process and piecewise segmentation with local extrema (turning points). Hence, Part1 (12/1/2019–2/4/2020), Part2 (2/4/2020–3/26/2020), Part3 (3/26/2020–6/10/2020), and Part4 (6/10/2020–6/30/2020) are constructed and each part has 44,

Table 11

T-test for the effect of EMIs.

		Part1	Part2	Part3	Part4	All Parts
Dataset3	Test Statistic	−3.865	−4.233	−1.773	−3.342	−2.752
	P-value	0.004	0.003	0.114	0.010	0.025
	Decision	Ha	Ha	Ho	Ha	Ha
Dataset4	Test Statistic	−3.342	−1.407	−3.267	N/A	−5.674
	P-value	0.010	0.197	0.011	N/A	0.004
	Decision	Ha	Ho	Ha	Ho	Ha

36, 52, and 14 trading days, respectively. The piecewise best feature subset of each time segment, which has the highest prediction accuracy among all feature subsets, was selected from the set of all non-empty power set of dataset3 and dataset4 over the whole testing period. Besides, performance metrics: Precision, Recall, F1, and AUC were measured and compared to the performance metrics of original XGB classifier as shown in [Table 10](#). The proposed PBFS selection method outperforms original XGB classifier over the whole testing period and each piecewise time segment. For the whole testing period, the PBFS prediction accuracy is 0.9041 and original XGB prediction accuracy is 0.8562 with dataset3. In each part, the PBFS prediction accuracies are 0.9091, 0.9730, 0.9434, and 0.9333, respectively. They are significantly higher than original XGB's predictions accuracies 0.8636, 0.8649, 0.8679, and 0.8667. Similarly, dataset4 also produces prediction accuracy of PBFS method: 0.9521 for the whole test period, and 0.9773 (part1), 0.9722 (Part2), and 1.000 (Part3) for each time segment, which are higher than 0.9315 for the whole test period, 0.9545 (Part1), 0.9444 (Part2), and 0.9808 (Part3) for each time segment using original XGB, respectively.

From the experiment of the PBFS on dataset3 and dataset4 as shown in [Table 10](#), the PBFS that is BFS of each time segment produces a higher classification accuracy with much fewer features than the original dataset of 10 input features. For instance, the ordinary XGB with dataset3 in time segment Part1 achieves the accuracy of 0.8636 with 10 technical indicators: Stok, Cci, Rsi, Willr, PCR, VIX, Mom, Ad, FGX, Macd as shown in [Table A.1](#) in [Appendix A](#). Meanwhile, the PBFS's accuracy 0.9091 is achieved only with 5 technical indicators: Stok, Cci, Willr, Rsi, and Mom as shown in [Table A.1](#) in [Appendix A](#). These experimental results, as shown in [Table A.1](#) through [Table A.4](#) in [Appendix A](#), empirically prove that the PBFS outperforms an ordinary XGB in both classification performance and the number of input features over all piecewise time segments. Hence, the proposed PBFS is more efficient with much fewer input features than ordinary XGB because the more features in a learning process generally causes overfitting, unnecessary memory consumption, expensive computational cost, and poor performance to a learning algorithm ([Yun et al., 2021](#)).

T-test results for the difference of average performance metrics also show that the PBFS has better performance in Part1, Part2, Part4, and all testing period with dataset3, and Part1, Part3, and all testing period with dataset4 than original XGB as shown in [Table 11](#).

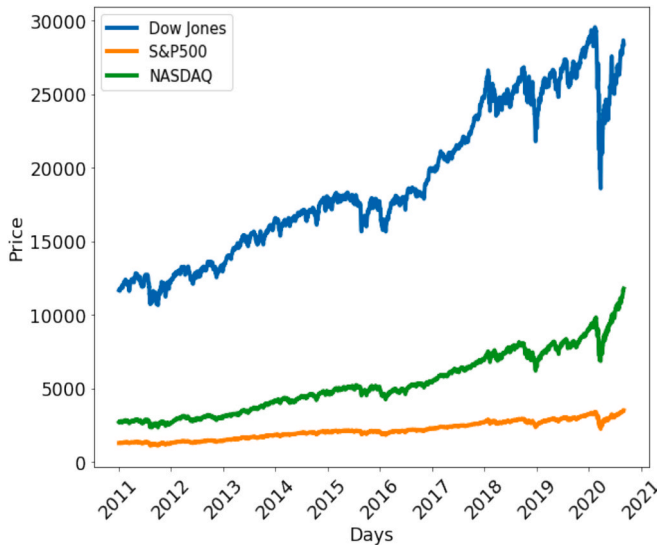
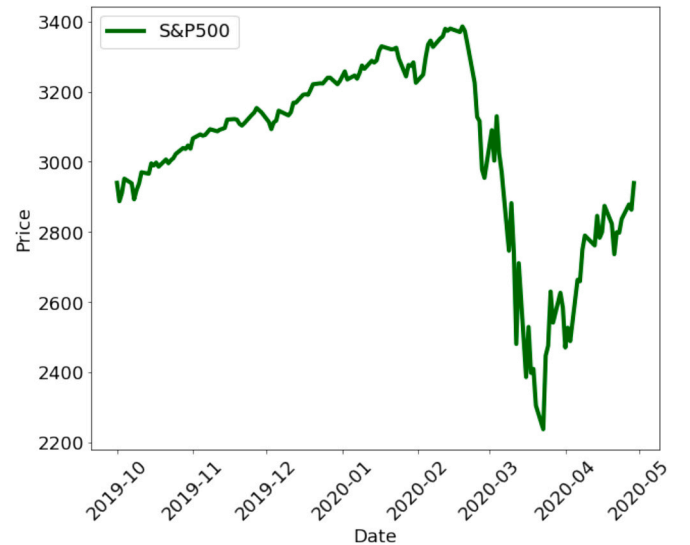
The feature-importance scores of dataset3 and dataset4 over the whole testing period were shown in [Table 12](#).

Over the whole testing period, PCR is the most important market indicator among the three EMIs. PCR ranked as the 5th important feature with 0.0745 importance score that follows Stok, Cci, Rsi, Willr by the PBFS method in dataset3, and the 4th important feature that follows Stok, Stod, and Bop in dataset4. The feature importance scores of each part of dataset3 and dataset4 were also measured and shown in [Appendix A](#). The proposed PBFS method generates BFSs with much fewer features for each part.

Table 12

Feature importance comparison of the whole test period.

	All Parts Best Accuracy	XGB 0.8562		PBFS 0.9041		3 EMIs only 0.6027	
Dataset3	FeatureImportance	Stok	0.4653	Stok	0.4937	PCR	0.4230
		Cci	0.0843	Cci	0.0953	FGX	0.2885
		Willr	0.0718	Rsi	0.0819	VIX	0.2884
		Rsi	0.0612	Willr	0.0791		
		PCR	0.0605	PCR	0.0745		
		VIX	0.0523	Mom	0.0641		
		Macd	0.0521	Macd	0.0620		
		Mom	0.0519	Ad	0.0493		
		Ad	0.0516				
		FGX	0.0489				
Dataset4	Best Accuracy	0.9315		0.9521		0.6027	
	FeatureImportance	Stok	0.4507	Stok	0.4786	PCR	0.4230
		Stod	0.1674	Stod	0.1863	FGX	0.2885
		Bop	0.1479	Bop	0.1551	VIX	0.2884
		Stochfh	0.0375	PCR	0.0445		
		Trange	0.0347	Stochfh	0.0391		
		PCR	0.0334	VIX	0.0377		
		Inphase	0.0312	FGX	0.0304		
		VIX	0.0301	Dosc	0.0284		
		FGX	0.0241				
		Mfi	0.0219				
		Dosc	0.0212				

**Fig. 8a.** Stock Market indices(2011 ~ 2020).**Fig. 8b.** S&P500 index (2019 ~ 2020).

4.6. Validation of the proposed methodology

The proposed BFS method is validated on different datasets under bull and bear market conditions. Three representative stock market indices: Dow Jones Index (DJI), S&P 500 Index (S&P), and Nasdaq Composite Index (Nasdaq) are used for the validation as shown in Fig. 8a and Fig. 8b.

Total 2123 trading days (2011/4/26 ~ 2019/10/1) are used for training data and 76 days (2019/10/1 ~ 2020/1/20) are used for bull market test data and 2217 trading days (2011/4/26 ~ 2020/2/15) are used for training data and 24 days (2020/2/15 ~ 2020/3/20) are used for Bear market test data.

Table 13 shows the performance metrics comparison of the proposed BFS and an ordinary XGB classifier on three datasets under bull and bear markets. The BFS method outperforms the XGB classifier on all three markets under both bull and bear markets. For example, the BFS method achieves the accuracies of 0.9342 and 0.9706 under bull and bear markets, respectively, but the XGB obtains 0.8947 and 0.9583 for the

accuracies for bull and bear markets, respectively.

To validate the proposed PBFS method, S&P500 Index is trained with 1159 trading days (2015/4/27 ~ 2019/11/30) and is tested with 146 trading days (2019/12/1 ~ 2020/6/30). The performance metrics of the PBFS is compared to the performance metrics of XGB in Table 14. The PBFS used 4 piecewise time segments and the mean accuracy of all 4 time segments is 0.966 that is higher than the accuracy of the XGB. Other classification performance metrics: Precision, Recall, F1 and AUC are also greater than the ordinary XGB.

4.7. Comparison to fixed-length time intervals

The performance of the PBFS method is compared to the performance over different fixed-length time intervals. To test the XGB classifier for the period between 12/1/2019 and 6/30/2020 (Total 146 trading days), piecewise time segments: Part1 (12/1/2019–2/4/2020), Part2 (2/4/2020–3/26/2020), Part3 (3/26/2020–6/10/2020), and Part4 (6/10/2020–6/30/2020) are constructed as used in Section 4.5 and tested with bimonthly, monthly, and biweekly time periods. Due to

Table 13

Performance comparison over different market and conditions.

Index	Method	Market	Accuracy	Precision	Recall	F1	AUC
DowJones	XGB	Bull	0.8947	0.9000	0.9375	0.9184	0.8795
		Bear	0.9583	0.8750	1.000	0.9333	0.9706
	BFS	Bull	0.9342	0.9388	0.9586	0.9485	0.9256
		Bear	0.9706	0.9286	1.000	0.9630	0.9762
S&P500	XGB	Bull	0.9605	0.9792	0.9592	0.9691	0.9611
		Bear	0.9310	0.9091	0.9091	0.9091	0.9268
	BFS	Bull	0.9868	1.000	0.9796	0.9897	0.9898
		Bear	0.9706	0.9375	1.000	0.9677	0.9737
Nasdaq	XGB	Bull	0.9211	0.9057	0.9796	0.9412	0.8972
		Bear	0.8750	1.000	0.7000	0.8235	0.8500
	BFS	Bull	0.9474	0.9412	0.9796	0.9600	0.9342
		Bear	0.9412	0.9444	0.9444	0.9444	0.9410

Table 14

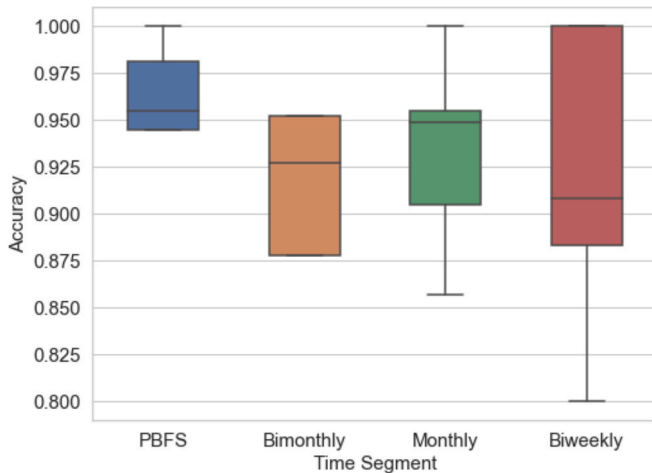
Performance comparison of the PBFS and XGB.

Method	Time Segment	Accuracy	Precision	Recall	F1	AUC
XGB	12/1 ~ 6/30	0.9315	0.9405	0.9405	0.9405	0.9299
PBFS	12/1 ~ 2/4	0.9545	0.9643	0.9643	0.9643	0.9509
	2/5 ~ 3/26	0.9444	1.0000	0.8750	0.9333	0.9375
	3/27 ~ 6/10	0.9808	1.0000	0.9677	0.9836	0.9839
	6/11 ~ 6/30	1.0000	1.0000	1.0000	1.0000	1.0000
	Mean	0.9699	0.9911	0.9518	0.9703	0.9681

Table 15

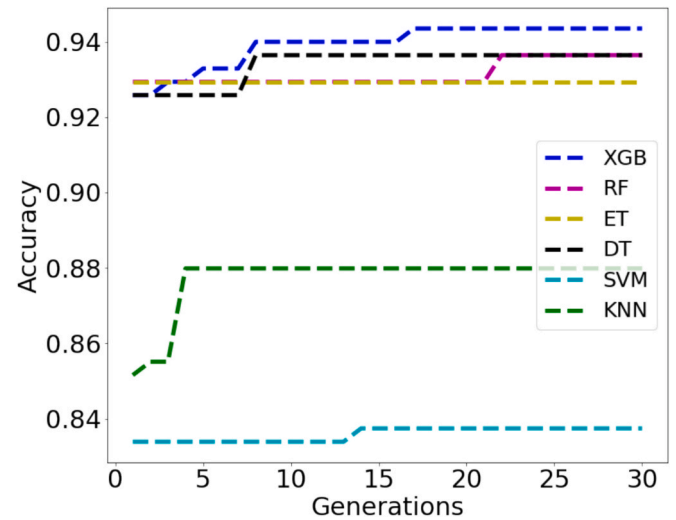
Performance comparison to different time intervals.

	PBFS	Bimonthly	Monthly	Biweekly
Average	0.9615	0.9191	0.9356	0.9236
S.D.	0.0182	0.0309	0.0447	0.0722

**Fig. 9.** Box-whisker plot over different time segments.

the different time intervals of each method, the average accuracy and the standard deviation of accuracies are calculated as shown in Table 15. The comparison result shows that the prediction accuracy of the PBFS outperforms the accuracy of fixed-interval time intervals. The standard deviation of the accuracy also shows that the PBFS method is more stable and accurate than other fixed-interval methods.

The distribution of prediction accuracies of each method is visualized as box-and-whisker plot as shown in Fig. 9. The accuracies of the PBFS are similar to each other and distributed are more narrowly

**Fig. 10.** Comparison of different ML algorithms.

clustered than fixed-interval time segments. This plot also supports that the PBFS method is more stable and reliable prediction methods than other time segments.

4.8. Comparison to other ML algorithms

To validate the robustness of the proposed XGB-based-EMIs methodology, the performances of five other ML models are compared to that of XGB's using Genetic Algorithm (GA). Five popular ML models: Extra Trees (ET), Decision Tree (DT), Random Forest (RF), SVM with RBF kernel (SVM), K-Nearest Neighbour (KNN) classifiers are tested on the S&P500 index price data of 2353 trading days from 2011 to 04-26 to 2020-08-28. The first 80 % (1882 days) of the 2353 trading days is used as training (the first 85 % of 1882 days) and validation (the last 15 % of 1882 days) datasets, and the last 20 % (471 days) of the 2353 days is used as the holdout dataset. For the parameters of GA, 30 generations, 0.5 crossover rate, and 0.375 for mutation rate are used. Fig. 10 shows that XGB has the highest prediction accuracy (0.9435) for validation dataset. DT and RF reach the same accuracy (0.9364), and ET (0.9293), KNN (0.8799) and SVM (0.8375) follow DT and RF in the order.

Besides the selected ML classifiers, Long-Short-Term-Memory Network (LSTM) model and Bidirectional LSTM (BiLSTM) model are compared to XGB for prediction accuracy and processing time on the holdout dataset. The built-in setting of hyperparameters is used for all ML and DL classifiers for pure comparison of nature behavior and original model performance. Fig. 11 shows the prediction accuracy and processing time of all compared models. As a result, XGB shows the highest prediction accuracy (93.0 %) than RF and ET (92.8 %) then DT (89.8 %), BiLSTM (84.1 %), LSTM (83.0 %) follow next. XGB has much

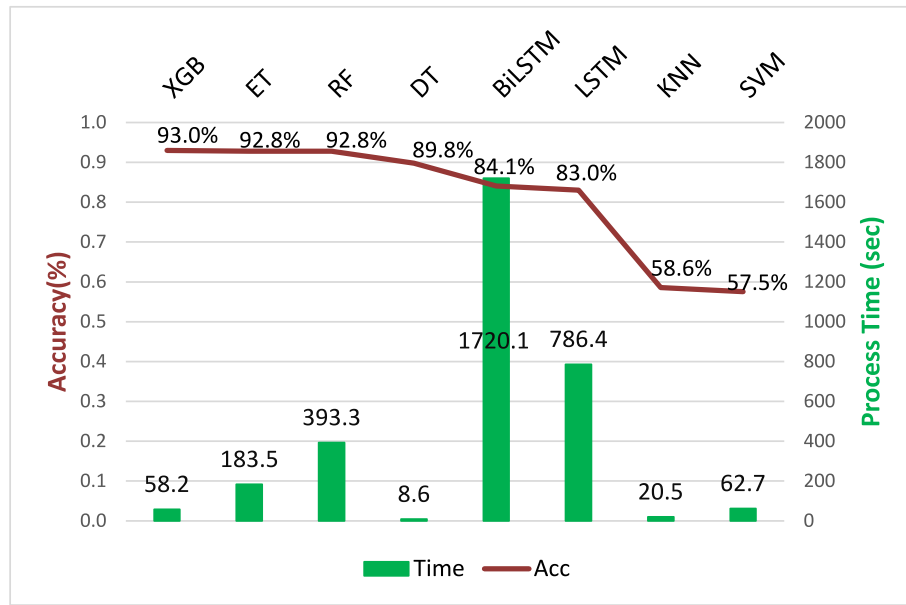


Fig. 11. Performance comparison of ML classifiers on holdout data.

faster process time (58.2 s) than ET, RF, LSTM, and BiLSTM.

The outperformance of XGB over other ML and DL algorithms can be ascribed to its superior scalability and parallel computing ability (Chen & Guestrin, 2016). The scalability of XGB is the ability to handle increasing amounts of data. Parallel computation is a way to distribute a computing task in parallel across Central Processing Unit (CPU) cores, which are individual processing units in the CPU. This scalability and computational parallelization of XGB supports fast calculations on large complex datasets.

In addition, XGB has an efficient regularization term that contains both L1 (Lasso) and L2 (Ridge) regularization as described in Formula (16).

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 + \alpha \sum |w_j| \quad (16)$$

where $\Omega(f)$ is a regularization term to control model complexity, f corresponds to an independent tree structure, and T, γ, w are the number of leaves in the tree, penalty for tree complexity, and leaf weights, respectively. λ is L2 (Ridge) regularization penalty for overfitting and α is L1 (Lasso) regularization penalty for promoting feature sparsity, respectively. With the combination of L1 and L2 in regularization term, XGB can efficiently avoid overfitting and promote sparsity in tree construction. By incorporating scalability with efficient regularization, XGBoost can maximize the prediction accuracy and minimize learning speed by exploiting every available hardware resource (Yun et al., 2021).

Table 16

Performance comparison to previous studies.

Dataset & Authors	Methods	Performance Metrics (%)				
		Accuracy	Precision	Recall	AUC	F1
Shanghai Stock Exchange Index (Wang, Xu, & Zheng, 2018)	Deep and Ensemble Learning	91.00	90.30	93.30	91.80	94.90
	BFS without EMIs	90.70	91.18	91.18	90.67	91.18
	BFS with EMIs	92.25	95.31	89.71	92.39	92.42
Bank of America Stock (Ampomah, Qin, & Nyame, 2020)	Ensemble Learning	84.63	84.29	85.82	85.05	92.80
	BFS without EMIs	92.63	90.16	95.93	92.58	92.96
	BFS with EMIs	92.92	90.22	96.15	92.87	93.26
Korean Stock Price Index (Chung and Shin, 2020)	Multi-channel CNN	73.74	N/A	N/A	N/A	N/A
	BFS without EMIs	94.15	93.81	95.50	94.02	94.64
	BFS with EMIs	95.61	96.36	95.50	95.62	95.93
CITIC Security Stock (Long, Chen, He, Wu, & Ren, 2020)	CNN + BiLSTM	75.89	N/A	N/A	N/A	73.10
	BFS without EMIs	93.55	92.86	92.86	93.49	92.86
	BFS with EMIs	93.33	96.30	89.66	93.21	92.86
S&P 500 Stock Index (Li and Pan, 2022)	Ensemble Deep Learning	N/A	60.00	75.00	N/A	66.67
	BFS without EMIs	85.71	66.67	100.00	90.00	80.00
	BFS with EMIs	95.81	97.27	94.69	95.87	95.96
MSFT Stock Price (Yang et al., 2022)	LASSO-LSTM	75.60	N/A	N/A	81.19	N/A
	BFS without EMIs	93.57	95.12	93.98	93.48	94.55
	BFS with EMIs	95.35	96.40	94.69	95.38	95.54
Hang Seng Stock Index (Zeng et al., 2023)	NSGA-II-RF	89.44	89.40	N/A	N/A	90.06
	BFS without EMIs	96.14	96.23	96.23	96.13	96.23
	BFS with EMIs	96.62	98.06	95.28	96.65	96.65
CHKJ Stock Price (Wang et al., 2025)	LSTM + Transformer	81.63	89.93	83.00	N/A	80.01
	BFS without EMIs	94.96	93.42	97.26	94.84	95.30
	BFS with EMIs	92.09	88.75	97.26	91.81	92.81

4.9. Discussion of the experimental result

As an experiment result, prediction accuracy and other performance metrics: Precision, Recall, AUC score, F1 increase with EMIs in all datasets. The higher accuracy the dataset has, the lower increment of accuracy it makes. The inclusion of EMIs not only increases prediction performance but also produces a parsimonious input feature set. In terms of the curse of dimensionality that high dimensional data set exacerbate model performance, the BFS of fewer input features improves model performance. The piecewise effect analysis by using piecewise time segment produces better predictability and interpretability for traders and investors in dealing with volatile, complex, and nonlinear stock prices as shown in Appendix A. The proposed PBFS method makes better prediction accuracy with fewer input features than original XGB classifier in each part. As a result, the effect of EMIs in stock price movement direction prediction is analysed both globally and locally.

As described in previous sections, this study intends to measure the effect and importance of the three exogenous market sentiment indicators in stock price direction prediction, but there are no direct and established methods to measure the importance and contribution degree of market sentiment indicators in stock price prediction. Hence, this study uses two relative but commonly-used straightforward metrics: model prediction accuracy and feature importance, which are changed depending on the permutation of input features in different datasets. For example, suppose there is an input feature f_1 . If it has the same importance score of 0.5 in two different datasets D1 and D2 that produce different prediction accuracies 90 % and 80 %, respectively. Then, f_1 in D1 can be considered more important than f_1 in D2 in terms of prediction accuracy and overall model performance. This accuracy-importance measures relieve the shortcomings of only using the feature importance of a single dataset, which can be disrupted by the relationship between features and outcome. Based on this rationale, different datasets that combine three EMIs and a variety of price-derived technical indicators are constructed and explored to reflect a relative importance of three EMIs and validate the contribution of them.

To find three EMIs' higher accuracy-importance, dataset2 through dataset4 are adopted from previous studies and compared to baseline dataset1 which consists of historical price values: Open, Low, High, Volume, and three EMIs as described in Section 3.6.2. The technical indicators in dataset2 through dataset4 are not only commonly used indicators but also proved to make good prediction accuracy in previous studies. The technical indicators of Dataset2 are adopted from a previous stock price prediction study (Yun et al., 2023), and they achieved optimal model performance in regression tasks. Dataset3 also has 7

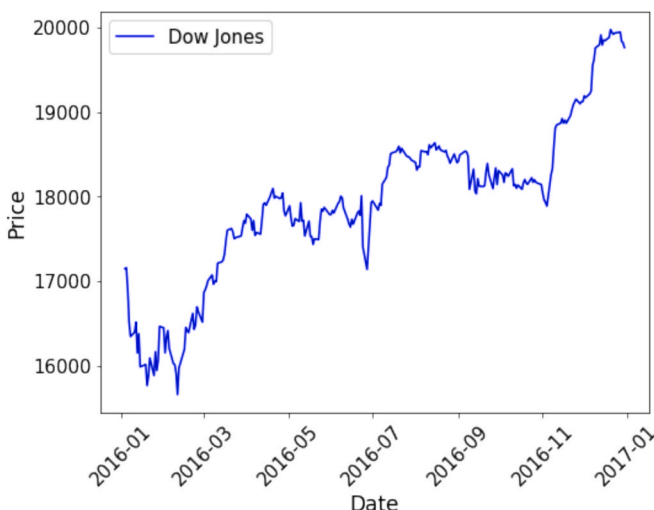


Fig. 12. Dow Jones Index (2016 /1 ~ 2026/12).

Table 17

Hyperparameters of DL models.

Epochs	Batch size	Learning rate	Dense layers	Dense nodes	Activation fuction
100	32	1.0E-05	1	20	Relu

Table 18

Performance comparison to other AI prediction models.

Dataset	Method	Accuracy	Precision	Recall	F1
Dow Jones(2011 ~ 2016)	Sentiment	0.7103	0.7266	0.7426	0.7345
	LSTM	0.8519	0.9171	0.7787	0.84225
	BiLSTM	0.7937	0.727	0.9155	0.81043
	EMIs	0.9363	0.9412	0.9412	0.9412

price-derived technical indicators. They are the most used technical indicators in previous stock price prediction studies (Basak et al., 2019; Chung & Shin, 2018; Chung & Shin 2020). Dataset 4 is borrowed from the previous study (Yun et al., 2021). They selected 33 optimal features out of 67 price-derived technical indicators and 4 historical stock prices (High, Low, Open, and Volume) for stock price direction prediction through a hybrid GA-XGB algorithm, then the best 7 price-derived technical indicators are selected out of the 33 optimal features. As the result of experiments over four different datasets, we can conclude that three EMIs make significant effect in stock price prediction. In Table 9 in section 4.4, Dataset1 accuracy increases from 54.67 % to 61.19 % with PCR, and Dataset2 accuracy increases from 78.47 % to 81.87 % with PCR, VIX, and FGX. Dataset3 accuracy also increases from 84.42 % to 86.97 % with PCR and VIX, and dataset4 accuracy increases from 95.18 % to 95.47 % with PCR. The accuracy increment in dataset4 is only 0.29 %, but the number of input features decreases from 7 to 3. Similarly, the PBFS selection of 4 different datasets validates the importance and contribution of three EMIs and the results are shown in Table 10 and Table A.1 through Table A.4 in Appendix A. This result implies that three market sentiment indicators make significant contributions to stock price direction prediction in terms of prediction accuracy and better interpretability with fewer input features.

5. Comparison to previous studies

The classification performance of the proposed BFS method were compared to recent stock price movement prediction studies and shown in Table16. The proposed BFS method outperformed all eight selected benchmark studies. In particular, the BFS with EMIs made the best performance metrics in datasets of Bank of America stock (Ampomah et al., 2020), Korean Stock Price Index (Chung & Shin, 2020), S&P 500 stock (Li & Pan, 2022), Microsoft stock (Yang et al., 2022), and Hang Seng Stock Index (Zeng et al., 2023).

6. Comparison to NLP-based sentiment analysis

To validate the effectiveness of using EMIs, the performance of the proposed BFS is compared to state-of-the-art stock prediction methods: NLP-based sentiment analysis, LSTM, and Bidirectional LSTM models. Dow Jones Index for the period of 2011/1/3 through 2016/12/30 (1510 trading days) is used for the comparison of all methods. Out of 1510 days of the dataset, the first 1258 days are used for training and validation, and the last 252 days are used for test data as shown in Fig. 12.

For sentiment analysis, VADER sentiment analysis tools from NLTK python library (Hutto & Gilbert, 2014) are used to convert Twitter sentiment message data about the stock market. After removing unnecessary characters and symbols, the twitter texts were converted to sentiment scores: compound, positive, negative, neutral. Then, they were used as input features with historical Open, High, Low, and Volume values of Dow Jones Index. LSTM model and Bidirectional LSTM

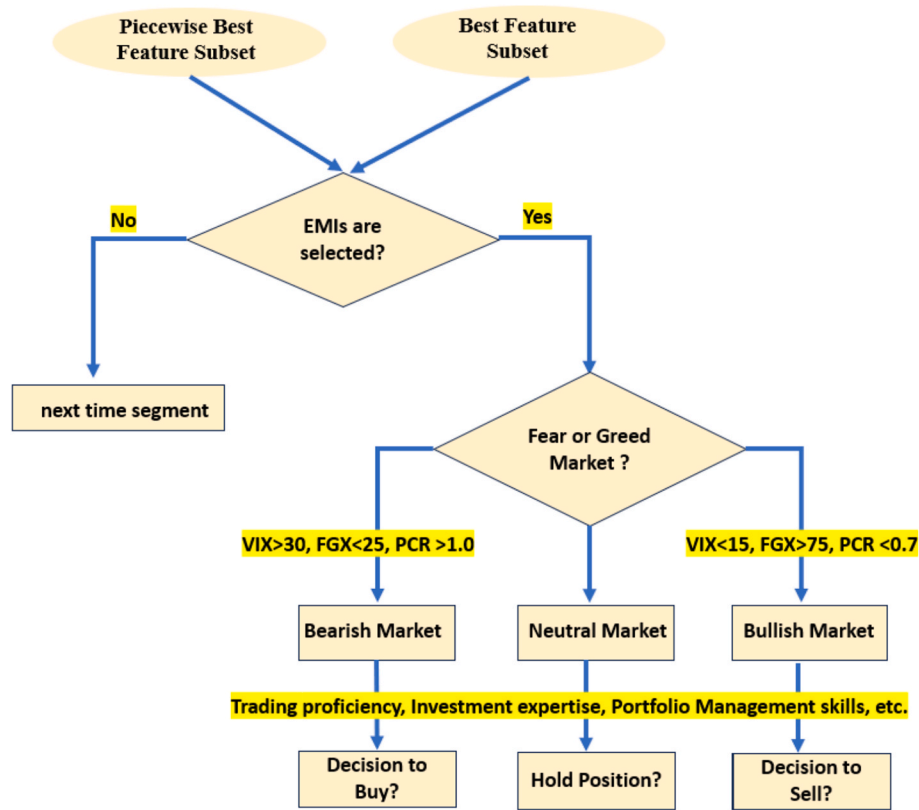


Fig. 13. Investment scenario with EMIs.

model are constructed with the following default hyperparameters as shown in Table 17.

The result in Table 18 shows that the proposed BFS using EMIs substantially outperformed other methods in the measurement of Accuracy, Precision, Recall, and F1. In particular, NLP-based sentiment analysis had the worst prediction performance.

7. Conclusion

The experimental result of this study reveals that correlation coefficients of EMIs with the outcome as well as with other price-derived technical indicators are not so important in stock price direction prediction. This nonlinear property of stock price data has been shown by many previous studies (Chung & Shin, 2018; Yun et al., 2021; Basak, Kar, Saha, Khaidem, & Dey, 2019). Despite weak or mediocre linear relationships, EMIs could make significant impact on the stock price movement because of the two important properties: temporality and collective behaviour of time series dataset (Yun et al., 2023). In other words, the prediction accuracy in stock price prediction greatly depends upon price timeline and other input features. However, the importance of these temporality and collective behaviour of time series stock price data were ignored by previous studies, but Yun et al. (2023) proposed a novel method: the BFS selection to capture the collective behaviour and the piecewise time segmentation to effectively reflect temporality of stock price data. In this study, as the continuation of the previous study, the BFS selection methods finds the best performing feature set that includes the selected EMIs by using the XGB algorithm, which reflects the collective behaviour of input features and significantly improves the prediction accuracy and interpretability with fewer input features.

One critical issue of stock price prediction studies is the employment of price-derived technical indicators alone. Price-derived technical indicators-based stock price prediction studies are criticized for ‘complete-subjectivity’ and ‘self-fulfilling prophecy’ problems (Murphy, 1999). As a remedial action to the problems of price-derived technical

indicators, this study constructs an input feature set of EMIs and price-derived technical indicators and uses the BFS and piecewise time segmentation method. Consequently, the Piecewise BFS method and the inclusion of EMIs certainly relieves the drawbacks of price-derived technical indicators. Further, combined with the piecewise time segmentation technique, this study captures the contribution and effect of EMIs in the stock price movement in small time segments, provides better piecewise interpretability, and it enables traders and investors to promptly response to constantly changing stock prices with the best accuracy of the input features.

This study explores the importance and effect of EMIs in stock price prediction using the best feature set and the piecewise best feature subset selection methods. To utilize the exogenous market sentiment indicators, there are many steps to transition from a theoretical study to a practical trading state. Although this study does not focus on the practical stock trading strategies, the following scenarios, as a plausible guideline, can be considered to implement the proposed methodology in real stock trading settings.

Once the BFS and the PBFS selection is completed, the following case actions can be recommended depending on each EMI’s importance and contribution in the BFS of each time segment as described in Fig. 13.

Case 1: EMIs are not selected.

In this case, the market can be considered neutral between fear and greed because of no significant effect of EMIs for a stock price movement. Hence, holding position is recommended for stock trading.

Case 2: EMIs are selected.

This case implies the EMIs may reflect the stock market movement. Two scenarios can be considered under this case.

- Scenario 1: Fear Market when $VIX > 30$, $FGX < 25$, or $PCR > 1.0$

The market sentiment may represent high degree of fear and gloom about the market. It can be interpreted as a counter-buy signal because the market sentiment is overly negative, and stocks may be oversold.

- Scenario 2: Greed Market when $VIX < 15$, $FGX > 75$, or $PCR < 0.7$

The market sentiment may imply high greed and overvalued market. Selling stock is recommended before the correction of the market.

While stock price prediction models can provide intuition and direction for the following trading strategies, there are many constraints to practice the prediction models in making real investment decisions because stock market is a complex system in which various forces and factors interplay convolutedly. Besides, market liquidity, data acquisition, trading speed, transaction fees, and constant model update are to be considered and solved for real-time stock trading. To bridge the gap between stock price prediction study and practical trading strategies and investment decisions, additional chapters of the research should follow this study.

The proposed BFS selection method with EMIs provides a better solution to the problem of price-derived technical indicators-based stock price prediction. However, there are several limitations of this research. First, the proposed methodology in this study has the problem of practicability and accessibility from the perspective of general users. Although the BFS with the XGB improves learning performance and prediction accuracy, the proposed methodology is still complex and difficult to use in a real stock market investment setting. This complexity and difficulty of the proposed method might limit its accessibility for practitioners without technical expertise in machine learning and feature selection techniques. Hence, the proposed methodology should be improved for practitioners to use easily in stock market prediction.

One way to improve the proposed methodology would be approached by using visualization. The construction of human-friendly visualization helps users understand the complex learning process and the selection of the BFS as well as increases the explainability of the outcome, on which appropriate trading strategies and transactions are to be developed. Hence, a human-friendly visualization of the BFS selection process can be explored in the following research. Secondly, this study adopted only three exogenous market indicators: VIX, Fear-greed index, and Put-call ratio as the substitute of price-derived technical indicators. As a complex system, stock market is affected by such various internal and external factors as economic situations, political issues, international relations, company-specific news, and many more. The stock market reflects the effect of those various factors with the change of market prices. However, the quantification of various important factors that can affect the market is generally difficult or sometimes impossible. In addition, it is not easy to obtain fair dataset including EMIs, that were used in this study: VIX, FGX, PCR. The difficulty of obtaining appropriate datasets is an obstacle to more comprehensive and rigorous research. As a result, only three EMIs with limited period were investigated in this research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. . Piecewise accuracy and feature importance

Table A1

Part1 Feature importance comparison.

Part1		XGB		PBFS		EMI	
Dataset3	Best Accuracy FeatureImportance	0.8636		0.9091		0.5455	
		Stok	0.4653	Stok	0.5980	PCR	0.4230
		Cci	0.0843	Cci	0.1218	FGX	0.2885
		Willr	0.0718	Willr	0.1042	VIX	0.2884
		Rsi	0.0612	Rsi	0.1025		
		PCR	0.0605	Mom	0.0735		
		VIX	0.0523				
		Macd	0.0521				
		Mom	0.0519				
		Ad	0.0516				
		FGX	0.0489				
Dataset4	Best Accuracy FeatureImportance	0.9545		0.9773		0.5455	
		Stok	0.4507	Stok	0.5182	PCR	0.4230
		Stod	0.1674	Stod	0.1715	FGX	0.2885
		Bop	0.1479	Bop	0.1524	VIX	0.2884
		Stochfh	0.0375	PCR	0.0459		
		Trange	0.0347	Inphase	0.0447		
		PCR	0.0334	Dosc	0.0342		
		Inphase	0.0312	Mfi	0.0331		
		VIX	0.0301				
		FGX	0.0241				
		Mfi	0.0219				

Table A2

Part2 Feature importance comparison.

Part2		XGB		PBFS		EMI	
Dataset3	Best Accuracy FeatureImportance	0.8649		0.9730		0.6757	
		Stok	0.4837	Stok	0.6023	PCR	0.4331
		Cci	0.0799	Mom	0.1071	VIX	0.2911
		Willr	0.0735	Ad	0.0996	FGX	0.2758
		Rsi	0.0675	PCR	0.0964		
		PCR	0.0645	VIX	0.0946		
		VIX	0.0556				
		Mom	0.0523				
		FGX	0.0446				
		Macd	0.0427				
Dataset4	Best Accuracy FeatureImportance	Ad	0.0357				
		0.9444		0.9730		0.6757	
		Stok	0.4532	Stok	0.5292	PCR	0.4331
		Stod	0.1738	Stod	0.1952	VIX	0.2911
		Bop	0.1353	Bop	0.1117	FGX	0.2758
		PCR	0.0360	PCR	0.0441		
		Stochfh	0.0346	FGX	0.0412		
		Dosc	0.0308	Inphase	0.0395		
		VIX	0.0284	Dosc	0.0391		
		FGX	0.0282				
		Trange	0.0281				
		Inphase	0.0267				
		Mfi	0.0249				

Table A3

Part3 Feature importance comparison.

Part3		XGB		PBFS		EMI	
Dataset3	Best Accuracy FeatureImportance	0.8679		0.9434			0.4528
		Stok	0.4640	Stok	0.5455	PCR	0.4344
		Cci	0.0811	Cci	0.1035	VIX	0.2884
		Rsi	0.0751	Willr	0.1030	FGX	0.2772
		Willr	0.0725	FGX	0.0992		
		PCR	0.0668	Mom	0.0854		
		VIX	0.0579	Ad	0.0634		
		Mom	0.0571				
		Ad	0.0456				
		FGX	0.0407				
Dataset4	Best Accuracy FeatureImportance	Macd	0.0392				
		0.9708		0.9722			0.4528
		Stok	0.4685	Stok	0.5447	PCR	0.4344
		Stod	0.1870	Bop	0.1781	VIX	0.2884
		Bop	0.1068	Dosc	0.0782	FGX	0.2772
		PCR	0.0396	VIX	0.0723		
		FGX	0.0351	PCR	0.0640		
		Trange	0.0327	FGX	0.0628		
		Inphase	0.0323				
		VIX	0.0317				
		Stochfh	0.0290				
		Mfi	0.0192				
		Dosc	0.0182				

Table A4
Part4 Feature importance comparison.

Part4		XGB		PBFS		EMI	
Dataset3	Best Accuracy	0.8667		0.9333		0.7333	
	FeatureImportance	Stok	0.4623	Stok	0.5379	PCR	0.4341
		Cci	0.0947	Cci	0.1005	VIX	0.2919
		Willr	0.0771	Rsi	0.0861	FGX	0.2739
		Rsi	0.0695	VIX	0.0785		
		PCR	0.0614	PCR	0.0720		
		VIX	0.0557	FGX	0.0664		
		Mom	0.0521	Macd	0.0586		
		FGX	0.0508				
		Macd	0.0397				
Dataset4	FeatureImportance	Ad	0.0367				
		1.0000		1.0000			0.7333
		Stok	0.4435	Stok	0.5432	PCR	0.4341
		Stod	0.1915	Stod	0.2119	VIX	0.2919
		Bop	0.1081	Bop	0.1136	FGX	0.2739
		PCR	0.0491	Inphase	0.0451		
		Inphase	0.0333	VIX	0.0434		
		Dosc	0.0333	PCR	0.0428		
		VIX	0.0325				
		Trange	0.0299				
		FGX	0.0284				
		Stochfh	0.0260				
		Mfi	0.0243				

Appendix B. . The correlation coefficient map of Dataset4

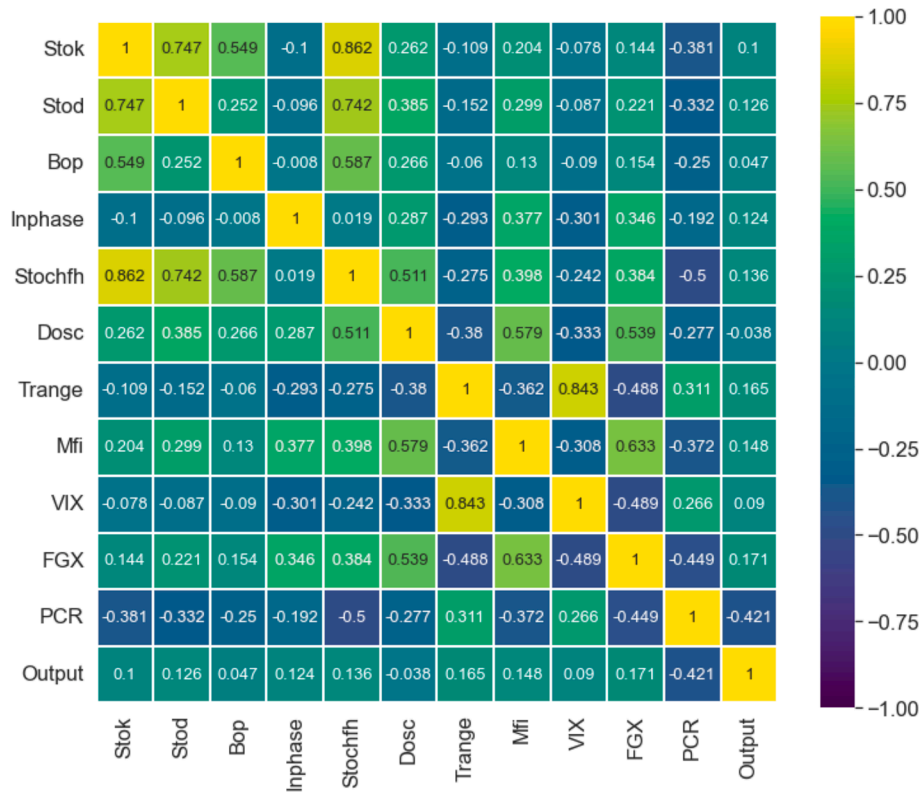


Fig. 14. Correlation coefficient map of Dataset 4.

Data availability

Data will be made available on request.

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